

# Product Switching and Young Firm Dynamics\*

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## Abstract

Firms learn which products they are good at—their product-specific expertise—by producing and selling them. We develop a general-equilibrium model of product switching over the firm life cycle in which firms differ in how precisely they learn: their ability to process information rises with managerial ability, age, and data. Better-informed firms drop weak products sooner and are more willing to search for new ones, so selection starts inside the firm. The model predicts that the link between expertise and product dropping is weaker in young firms and stronger in better-managed, more data-intensive firms. We confirm these predictions using a new measure of expertise from Census data on U.S. manufacturing firms, linked to the Management and Organizational Practices Survey. In the calibrated model, subsidizing startups' managerial investment lowers survival but raises consumption by 0.5 percent net of cost via better production decisions and product selection.

**JEL Code:** L11, L21, L25, L60, O31

**Keywords:** product add and drop, product switching, young firm dynamics, firm growth, resource allocation, aggregate productivity

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# 1 Introduction

Firms must learn not only how good they are, but also what they are good at. In the canonical model of [Jovanovic \(1982\)](#), firms learn how efficient they are, and this learning determines which firms grow and which exit. Yet a firm may be good at making and selling some products and poor at others, and it learns which is which mainly through experience. Selection therefore starts inside the firm: just as workers discover their comparative advantage by working in different occupations, firms discover theirs by switching products.<sup>1</sup> In this paper, we develop a theory of product switching over the firm life cycle in which firms learn about their product-specific expertise—how well they can make and sell each product—and we test its predictions using Census data on U.S. manufacturing firms and their products.

The central idea is that firms differ in how precisely they learn, and that this precision improves with age. Younger firms with less experience, and firms with less knowledge of how to interpret data or less available data, form noisier beliefs, keep weak products—those they are poor at—too long, and sometimes drop strong ones. Better-informed firms drop weak products sooner and keep searching for better ones, so learning amplifies the up-or-out dynamics of young firms ([Haltiwanger et al., 2013](#)). Our quantitative analysis shows that better information raises firm performance and aggregate consumption even as fewer firms survive, so survival understates its value.

Product switching is economically important. In our data, 65 percent of continuing U.S. manufacturing firms add or drop a product over a five-year interval, and these firms account for 91 percent of the output of continuing firms, consistent with [Bernard et al. \(2010\)](#). Many of these firms replace products rather than only adding or only dropping them: 34 percent of continuing firms both add and drop products within the same interval. Adding and dropping products accounts for 56 percent of growth among firms aged five or less and 44 percent among older firms.<sup>2</sup> How well firms choose their

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<sup>1</sup>On how workers learn about their ability, comparative advantage, and job-match quality, see [Jovanovic \(1979\)](#), [Papageorgiou \(2014\)](#), [Groes et al. \(2015\)](#), and [Baley et al. \(2022\)](#).

<sup>2</sup>We measure growth as the change in the shipment-weighted average of log real product shipments.

products therefore shapes how they grow over the life cycle and how quickly resources move toward their best uses.

Our first contribution is theoretical. Canonical models of firm learning study learning about a firm-level characteristic (Jovanovic, 1982; Arkolakis et al., 2018), while in models of product switching, firms either observe their product-specific profitability (Bernard et al., 2010) or learn about demand common to all of their products (Timo-shenko, 2015). We develop a general-equilibrium model with free entry in which firms can operate several products at once, learn about each product separately, choose which products to keep learning about, and differ in how precisely they learn. Each firm has a permanent but unobserved expertise in each product and learns about it from noisy realized demand and an additional signal whose precision—its information-processing ability—rises with managerial ability, experience, and accumulated data (Farboodi and Veldkamp, 2026; Baley and Veldkamp, 2025). Each period, the firm decides which products to keep and whether to search for new ones, and it exits when it drops all of them. Product tenure and information processing play different roles: a mature firm that adds a new product starts from the same prior as a startup but can learn faster.

The theory links information processing to both production and selection. First, less precise information strengthens the shrinkage of beliefs toward the prior, so relative to a benchmark with known expertise, firms in expectation underproduce products with sufficiently high expertise and overproduce those with sufficiently low expertise. Second, firms keep uncertain products partly for the option value of learning more, and better information changes this value because firms know more about their existing products and can evaluate new ones more quickly. Under conditions we characterize, better information raises both the drop threshold and the search threshold, so better-informed firms are more selective about which products they keep and more willing to search for new ones. Under related conditions, higher productivity also raises the search threshold but lowers the drop threshold, because it makes weak products profitable enough to keep. Both push firms to keep searching for better products, but only better information makes them drop weak products sooner. The model predicts that selection

on expertise is weaker among younger firms and among firms with lower managerial ability or less accumulated data.

Our second contribution is empirical. Prior work shows that product switching is pervasive and shapes firm growth (Bernard et al., 2010; Argente et al., 2018); our new empirical measure of product-specific expertise lets us ask whether firms keep the products they are good at. Using the Census of Manufactures from 1977 to 2012, we estimate expertise as the persistent firm–product component of real shipments, as motivated by the model’s demand structure, controlling for product–year conditions, firm–product tenure, and time-varying firm characteristics, including the number of products a firm makes and its total assets. Conditional on current shipments, a one-standard-deviation increase in estimated expertise is associated with a 26-percentage-point decline in the probability of dropping a product over the next five years among mature firms, compared with 14 percentage points among firms aged five or less. The age difference is robust to using continuous firm age and to estimating expertise from the same number of observations for every firm–product pair. Using the Management and Organizational Practices Survey (MOPS), we further show that selection on expertise is stronger among firms with higher structured management scores, greater data availability, and more intensive use of data in decision making.

Our third contribution is quantitative. Prior work shows that imperfect information distorts how firms allocate inputs (David et al., 2016); we show that it also shapes which products firms keep and which firms survive. We calibrate the model to U.S. manufacturing, using the product-dropping regressions to discipline information-processing ability. The calibrated model generates up-or-out dynamics: among the most productive firms, better-managed firms are less likely to survive to age 20 but more likely to become large. We then study a subsidy to startups’ managerial investment that costs 0.1 percent of consumption. Following the same entrants with and without the subsidy at baseline prices, survival to age 20 falls from 20.1 to 18.0 percent, but cumulative real sales per entrant, counting zero sales after exit, rise by 2.8 percent. Two-thirds of this gain comes from production adjustments within existing portfolios and two-fifths

from changes in which products firms operate, more than offsetting a small loss from lower survival. In general equilibrium, consumption rises by 0.54 percent net of the program’s cost, even though the number of firms falls by 7.7 percent. Better information thus leads to fewer but better firms with better products.<sup>3</sup>

Finally, we characterize welfare. Holding fixed startups’ investment in productivity and managerial ability before entry, the competitive equilibrium can be replicated by a restricted planner who faces the same information as firms and retains the equilibrium’s value-function restrictions on product-switching, startup, and entry decisions. The dispersion that learning creates is therefore not, by itself, a sign of inefficiency. Instead, knowledge spillovers across cohorts provide a rationale for intervention: startups do not take into account that part of the productivity and managerial know-how they build passes on to later entrants (Romer, 1990; Jones, 1995; Sampson, 2016).

## Related Literature

First, we contribute to research on firm dynamics and learning. Canonical models study selection on a firm-level state (Jovanovic, 1982; Hopenhayn, 1992; Luttmer, 2007), and Arkolakis et al. (2018) link demand learning to firm growth. Empirical work highlights the role of demand in selection and in the slow growth of new businesses (Foster et al., 2008, 2016; Eslava et al., 2024), shows that firm size largely reflects product appeal and scope (Hottman et al., 2016), and finds that much heterogeneity is present at birth (Sterk et al., 2021). We combine such heterogeneity with product-level learning, so that firm-level selection emerges from product-level decisions.

Second, we contribute to research on product switching and firm growth. Product turnover is pervasive and shapes firm growth (Bernard et al., 2010; Broda and Weinstein, 2010; Argente et al., 2018, 2024; Berlingieri et al., 2026), and firms tend to add products that share inputs with their existing ones (Boehm et al., 2022).<sup>4</sup> Models of

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<sup>3</sup>The phrase is from Ates and Saffie (2021), who show that tighter credit during sudden stops leads to fewer but more productive entrants. Here, selection instead reflects better information after entry.

<sup>4</sup>In trade models, competition shifts production toward the products firms know they are best at

innovation, reallocation, and distortions also treat firms as collections of product lines (Klette and Kortum, 2004; Akcigit and Kerr, 2018; Acemoglu et al., 2018; Fattal Jaef, 2018). Close antecedents are Mitchell (2000), in which information links the tasks a firm undertakes and limits its scope, and Timoshenko (2015), in which firms learn about demand common to their products in a destination and adjust scope along a known ranking of product-specific efficiencies. We instead study learning about expertise in each product and empirically test its predictions.

Third, we relate to work on information and resource allocation. Imperfect information affects how firms allocate inputs and lowers aggregate productivity (David et al., 2016; David and Venkateswaran, 2019). Management practices are associated with productivity (Bloom et al., 2019) and with more accurate forecasts (Bloom et al., 2025), and data can improve firms' decisions (Begenau et al., 2018; Farboodi et al., 2019; Farboodi and Veldkamp, 2026; Baley and Veldkamp, 2025). Asriyan and Kohlhas (2026) show that data help firms choose their scale and which products to offer; we show that differences in how precisely firms learn over the life cycle also shape which products they keep, when they search for new ones, and whether they exit. In our model, the dispersion that learning creates need not reflect inefficiency, as Kehrig and Vincent (2026) emphasize in a different setting, and the case for policy comes from knowledge spillovers across cohorts.<sup>5</sup>

Finally, our mechanism complements explanations of young-firm dynamics based on financing constraints (Evans and Jovanovic, 1989; Cooley and Quadrini, 2001; Schmalz et al., 2017), customer accumulation (Gourio and Rudanko, 2014; Foster et al., 2016), technology diffusion (Akcigit and Ates, 2023; Jo and Kim, 2026), and workers' uncertainty about young employers (Kim, 2025). We study a different friction: firms do not know which products they are good at, and they learn this more precisely as they age.

The rest of the paper proceeds as follows. Section 2 presents the model and its theoretical predictions.

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<sup>5</sup>Our welfare analysis also relates to work on knowledge accumulation and diffusion (Romer, 1990; Jones, 1995; Perla and Tonetti, 2014; Sampson, 2016; Buera and Oberfield, 2020) and on the transfer of managerial and entrepreneurial knowledge (Giorcelli, 2019; Guiso et al., 2021).

oretical results. Section 3 describes the data, constructs our measure of expertise, and tests the model's predictions. Section 4 calibrates the model and studies selection within and across firms, the managerial-investment subsidy, and welfare. Section 5 concludes.

## 2 Theoretical Framework

This section presents the model. Firms learn about their expertise in each product, and their information-processing ability depends on managerial ability, firm age, and accumulated data, so firms learn at different rates. The model builds on learning about firm fundamentals (Jovanovic, 1982; Kim, 2025) and on product switching (Bernard et al., 2010), and the role of information processing is motivated by work on the data economy (Farboodi et al., 2019; Farboodi and Veldkamp, 2026; Baley and Veldkamp, 2025). We first describe the environment, consumers, and firms, then potential startups and the equilibrium, and finally derive the model's implications.

### 2.1 Environment

The economy contains a finite set of products  $p \in \{1, \dots, N\}$  and a continuum of firms indexed by  $i$ . Product markets are monopolistically competitive, with firms producing differentiated goods within each market. Time is discrete, indexed by  $t$ .

### 2.2 Consumers

A representative household supplies its labor endowment  $\bar{L}$  inelastically and owns all firms. It maximizes

$$\sum_{t=0}^{\infty} \beta^t \log \bar{C}_t,$$

subject to the period budget constraint  $\sum_{p=1}^N \int_{i \in \Omega_{pt}} p_{ipt} c_{ipt} di + w_t \mathcal{F}_t^e \leq w_t \bar{L} + \Pi_t$ , where  $\beta \in (0, 1)$  is the discount factor. Here,  $\Omega_{pt}$  is the set of firms operating in product market  $p$ ,  $p_{ipt}$  and  $c_{ipt}$  are the price and consumption of firm  $i$ 's good in market  $p$ ,  $w_t$  is the wage,

and  $\Pi_t$  denotes aggregate profits before entry costs. Aggregate entry costs, including pre-entry capability investments, are  $\mathcal{F}_t^e$  in labor units.

The aggregate consumption index,  $\bar{C}_t$ , and product-level consumption indices,  $C_{pt}$ , are defined by nested constant elasticity of substitution (CES) aggregators:

$$\bar{C}_t = \left( \sum_{p=1}^N C_{pt}^{\frac{\nu-1}{\nu}} \right)^{\frac{\nu}{\nu-1}}, \quad C_{pt} = \left[ \int_{i \in \Omega_{pt}} (\theta_{ipt} c_{ipt})^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}}.$$

The elasticities of substitution across products and differentiated goods within each market are  $\nu$  and  $\eta$ , respectively, with  $1 < \nu < \eta$ . The firm-product demand shifter  $\theta_{ipt}$  combines permanent product-specific expertise,  $\theta_{ip}$ , with a transitory demand shock, as specified below. Expenditure minimization yields demand for each good:<sup>6</sup>

$$c_{ipt} = P_{pt}^\eta C_{pt} \theta_{ipt}^{\eta-1} p_{ipt}^{-\eta}. \quad (2.1)$$

The corresponding aggregate and product-level price indices are

$$\bar{P}_t = \left( \sum_{p=1}^N P_{pt}^{1-\nu} \right)^{\frac{1}{1-\nu}}, \quad P_{pt} = \left[ \int_{i \in \Omega_{pt}} \theta_{ipt}^{\eta-1} p_{ipt}^{1-\eta} di \right]^{\frac{1}{1-\eta}}.$$

## 2.3 Firms

Before entry, potential startups invest in productivity and managerial capabilities, summarized by  $z_0$  and  $m_0$ , respectively. Higher  $z_0$  raises productivity, while higher  $m_0$  improves information-processing ability and accelerates learning about product-specific expertise. We return to these pre-entry choices in Section 2.4.

Firms are born with two permanent fundamental types: productive potential  $\varphi_i$  and managerial ability  $m_i$ . Productive potential determines productivity  $z_i \equiv z(\varphi_i; z_0)$  and potential product scope  $n_i \equiv n(\varphi_i)$ , the maximum number of product lines a firm can operate simultaneously. Productivity is strictly increasing in  $\varphi_i$ , while potential scope

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<sup>6</sup>The inverse demand function is  $p_{ipt} = P_{pt}^{1/\eta} \theta_{ipt}^{(\eta-1)/\eta} c_{ipt}^{-1/\eta}$ .

is nondecreasing. Managerial ability  $m_i$  may be correlated with  $\varphi_i$  and is drawn from a distribution shifted by  $m_0$ . In each period, firms decide for each product line whether to add or drop a product and choose quantities for all operating products.

### 2.3.1 Firm-Product Expertise and Learning

In addition to its fundamental types  $(\varphi_i, m_i)$ , firm  $i$  is born with time-invariant expertise  $\theta_{ip}$  in each product  $p$ , drawn from a lognormal distribution,  $\log \theta_{ip} \sim N(\theta_0, \sigma_0^2)$ . Expertise captures how well the firm is suited to meeting consumer needs in a particular product market, including preferences for product quality, design, and functionality. We model expertise as a demand-side attribute that raises consumers' valuation of the firm's good and its profitability.<sup>7</sup>

Firms do not observe  $\theta_{ip}$  directly. After production and sales, they observe a noisy signal inferred from the market-clearing price induced by realized demand:

$$\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt},$$

where  $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$  is an i.i.d. demand shock. Learning about  $\theta_{ip}$  follows Bayesian updating based on  $\log \theta_{ipt}$  and an additional signal  $\log s_{ipt}$  obtained by combining realized market performance with other information, such as knowledge of customer tastes and market conditions. Firms with greater experience, data, and managerial ability can extract a more precise signal from this information (Farboodi et al., 2019; Farboodi and Veldkamp, 2026; Baley and Veldkamp, 2025; Bloom et al., 2025).<sup>8</sup> We model this additional signal as

$$\log s_{ipt} = \log \theta_{ip} + v_{ipt}, \quad v_{ipt} \sim N\left(0, \frac{\sigma_\varepsilon^2}{\psi_{it}}\right). \quad (2.2)$$

We define  $\psi_{it}$  as the firm's information-processing precision, with higher  $\psi_{it}$  implying

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<sup>7</sup>Alternatively, expertise can be modeled on the production side as product-specific productivity.

<sup>8</sup>An alternative interpretation is that firms use their information about customer tastes and market conditions to filter transitory demand fluctuations from realized market performance.

a more precise additional signal. A firm's ability to process observable information depends on its managerial ability  $m_i$ , experience (captured by firm age  $a_{it}$ ), and accumulated data  $D_{it}$ , which depend on  $a_{it}$ , potential product scope  $n_i$ , and potentially other data sources. We summarize these factors by  $\psi_{it} \equiv \psi(m_i, a_{it}, D_{it}; \varphi_i)$ , with  $\partial\psi_{it}/\partial x \geq 0$  for  $x \in \{m_i, a_{it}, D_{it}\}$ .<sup>9</sup> Conditional on  $(\varphi_i, m_i)$ ,  $a_{it}$  determines the evolution of  $\psi_{it}$ .

Given this structure, firm  $i$  enters market  $p$  with prior  $\log \theta_{ip} \sim N(\theta_0, \sigma_0^2)$ . After production and sales, it observes the market outcome and additional signal and updates its beliefs about  $\theta_{ip}$ . After  $\tau_{ipt} \geq 1$  production periods, including  $t$ , its information set is  $\mathcal{I}_{ipt} \equiv \{\theta_{ipt}, s_{ipt}\}_{\tilde{t}=t-\tau_{ipt}+1}^t$ . This yields the posterior mean and variance of  $\log \theta_{ip}$ :

$$\hat{\theta}_{ipt} = \hat{\sigma}_{ipt}^2 \left[ \frac{\theta_0}{\sigma_0^2} + \frac{\tau_{ipt}}{\sigma_\varepsilon^2} \left( \frac{1}{\tau_{ipt}} \sum_{\tilde{t}=t-\tau_{ipt}+1}^t \log \theta_{ipt\tilde{t}} \right) + \frac{\sum_{\tilde{t}=t-\tau_{ipt}+1}^t \psi_{i\tilde{t}} \log s_{ipt\tilde{t}}}{\sigma_\varepsilon^2} \right], \quad (2.3)$$

$$\hat{\sigma}_{ipt}^2 = \left[ \frac{1}{\sigma_0^2} + \frac{\tau_{ipt}}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}=t-\tau_{ipt}+1}^t \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \right]^{-1} = \left[ \frac{1}{\hat{\sigma}_{ip,t-1}^2} + \frac{1 + \psi_{it}}{\sigma_\varepsilon^2} \right]^{-1}. \quad (2.4)$$

The posterior mean is a weighted average of the prior mean  $\theta_0$ , the average observed outcome, and additional signals, with weights determined by firm-product tenure  $\tau_{ipt}$  and the history of information-processing ability  $\psi_{i\tilde{t}}$ . A central feature of the model is the distinction between firm-product tenure and information-processing ability; Lemma 1 shows how the latter shapes posterior beliefs at a given tenure. The posterior also determines the predictive distribution of  $\theta_{ip,t+1}$ .

Since posterior variance is determined by firm-product tenure, firm age, and fundamental types, posterior beliefs can be tracked by  $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i, m_i)$ . We retain  $\psi_{it}$  explicitly to emphasize the role of information-processing ability. All firms start with the common prior  $(\hat{\theta}_{ipt}^0, (\hat{\sigma}_{ipt}^0)^2) \equiv (\theta_0, \sigma_0^2)$  at tenure  $\tau_{ipt}^0 = 0$ .

Based on this, we characterize how posterior beliefs about product-specific expertise  $\theta_{ip}$  vary with firms' information-processing ability at a given firm-product tenure.

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<sup>9</sup>The cumulative number of products dropped can serve as another proxy for firm experience. We abstract from this variable because it introduces dependencies across product lines, reducing tractability.

**Lemma 1** (Biased Learning). *For fixed  $\tau \geq 1$ , lower information-processing ability  $\psi_{it}$  increases posterior variance and magnifies, in expectation, underestimation of  $\log \theta_{ip}$  when  $\log \theta_{ip} > \theta_0$  and overestimation when  $\log \theta_{ip} < \theta_0$ , holding all else equal.*<sup>10</sup>

*Proof.* See Appendix A.1. □

Lemma 1 shows that lower information-processing ability leaves posterior beliefs more strongly influenced by the prior mean  $\theta_0$ . Noisier additional signals receive less weight, magnifying average underestimation of high product-specific expertise and overestimation of low expertise. These deviations reflect Bayesian shrinkage toward the prior mean under less precise information.

Firm-product tenure and firm age affect learning through distinct channels. Longer tenure  $\tau$  provides more product-specific observations, whereas greater information-processing ability  $\psi_{it}$  yields more precise additional signals. Conditional on  $(\varphi_i, m_i)$ ,  $\psi_{it}$  increases with firm age. Thus, when young and mature firms begin operating a product with the same prior, the mature firm can learn faster, with lower posterior variance and weaker shrinkage toward the prior at the same product tenure. Differences in managerial ability  $m_i$  generate learning heterogeneity among firms of the same age and product tenure. These differences can lead to systematically different product-switching decisions over the firm life cycle.

Lastly, we characterize the transition of posterior beliefs conditional on continued production. Under our functional-form assumption,  $\psi_{it}$  is determined by  $(a_{it}, \varphi_i, m_i)$ , so both  $\hat{\sigma}_{ipt}^2$  and  $\psi_{i,t+1}$  are known given the current state. Continued production implies  $\tau_{ip,t+1} = \tau_{ipt} + 1$  and  $a_{i,t+1} = a_{it} + 1$ . After observing the market outcome and additional signal in period  $t + 1$ , Bayesian updating gives  $\hat{\sigma}_{ip,t+1}^2 = (1/\hat{\sigma}_{ipt}^2 + (1 + \psi_{i,t+1})/\sigma_\varepsilon^2)^{-1}$  and  $\hat{\theta}_{ip,t+1} = \hat{\sigma}_{ip,t+1}^2 \left( \hat{\theta}_{ipt}/\hat{\sigma}_{ipt}^2 + \log \theta_{ip,t+1}/\sigma_\varepsilon^2 + \psi_{i,t+1} \log s_{ip,t+1}/\sigma_\varepsilon^2 \right)$ .

Integrating over  $\log \theta_{ip}$  using the current posterior and over the independent signal errors yields a Gaussian transition for the posterior mean.<sup>11</sup> The transition kernel  $g_{it}(\cdot |$

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<sup>10</sup>Expectations average over all potential signal histories within the observation window  $\tau$ , without conditioning on endogenous product retention.

<sup>11</sup>The next posterior mean is conditionally Gaussian because it is an affine function of the jointly

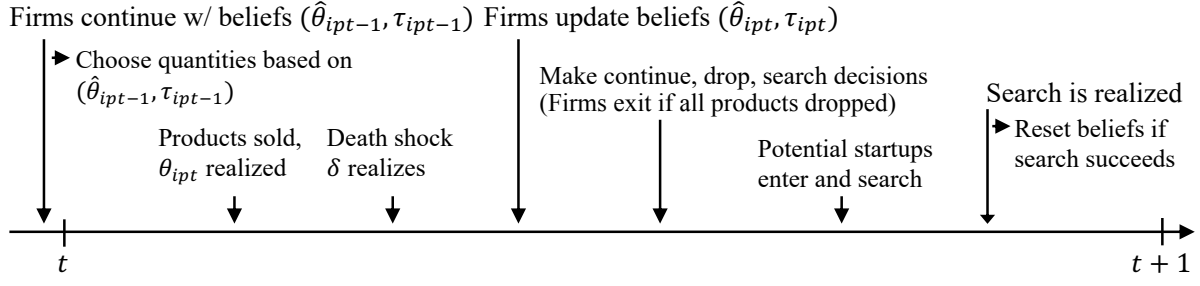


Figure 1: Timeline of the model

$\hat{\theta}_{ipt}, \tau_{ipt}) \equiv g(\cdot \mid \hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i, m_i)$  is

$$g_{it}(\hat{\theta}_{ip,t+1}, \tau_{ip,t+1} \mid \hat{\theta}_{ipt}, \tau_{ipt}) = \phi\left(\hat{\theta}_{ip,t+1}; \hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2\right) \mathbf{1}\{\tau_{ip,t+1} = \tau_{ipt} + 1\}, \quad (2.5)$$

where  $\phi$  denotes the normal density, parameterized by its mean and variance. The transition variance equals the reduction in posterior uncertainty, while the expected next posterior mean equals its current value. Hence, the posterior-mean process is a martingale under the firm's predictive beliefs.

### 2.3.2 Timing of the Model

Figure 1 illustrates the timing. Firms begin period  $t$  with posterior beliefs about  $\theta_{ip}$  and predetermined quantities, both based on  $\mathcal{I}_{ipt-1}$ , prior to the realization of  $\theta_{ipt}$ .<sup>12</sup> Demand and market-clearing prices are then realized given  $\theta_{ipt}$ , followed by exogenous death with probability  $\delta$ . Remaining firms update their beliefs using realized outcomes and additional signals and decide which products to retain or drop. If all active products are dropped, the firm exits before search; otherwise, it can search for new products.

Potential startups make entry decisions concurrently with incumbents' search decisions. Search outcomes and entrants' product assignments are realized at the end of the period. Continuing firms choose next-period quantities using updated beliefs for

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Gaussian log  $\theta_{ip}$  and signal errors. By the law of total variance, its conditional variance equals the current posterior variance minus the next posterior variance, which is known given the current state.

<sup>12</sup>Entrants begin with the prior  $(\hat{\theta}_{ip}^0, (\hat{\sigma}_{ip}^0)^2)$  and observe  $m_i$  after initial production, before their first belief update.

retained products, while beliefs for new products reset to the initial prior upon successful switching into a new product market. We define the value function after the death shock and belief updating, before product-retention and search decisions.

### 2.3.3 Production

Firms produce each product using linear technology with labor as the only input,  $y_{ipt} = z_i \ell_{ipt}$ , where  $z_i = z(\varphi_i; z_0)$ . Each operating product incurs a fixed cost  $f^o$  in labor units. Quantities are chosen using posterior beliefs about  $\theta_{ip}$  before  $\theta_{ipt}$  is realized. In this subsection, expectations are conditional on the current state  $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i, m_i)$ , and firms take  $(P_{pt+1}, C_{pt+1}, w_{t+1})$  as given and known. For a product continuing into period  $t + 1$ , the firm maximizes expected per-period profit:

$$\bar{\pi}_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \max_{y_{ipt+1} \geq 0} \left\{ \mathbb{E}_{\theta_{ipt+1}} [p_{ipt+1}(y_{ipt+1}, \theta_{ipt+1})] y_{ipt+1} - w_{t+1} \left( \frac{y_{ipt+1}}{z_i} + f^o \right) \right\}, \quad (2.6)$$

subject to demand (2.1). The optimal quantity is

$$y_{ipt+1} = \left( \frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} \left[ \mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \right]^\eta \left( \frac{z_i}{w_{t+1}} \right)^\eta, \quad (2.7)$$

where  $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \mathcal{B}(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i, m_i)$  is the conditional expectation of  $\theta_{ipt+1}^{(\eta-1)/\eta}$ :

$$\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) = \exp \left\{ \frac{\eta - 1}{\eta} \hat{\theta}_{ipt} + \frac{1}{2} \left( \frac{\eta - 1}{\eta} \right)^2 (\hat{\sigma}_{ipt}^2 + \sigma_\varepsilon^2) \right\}. \quad (2.8)$$

Substituting (2.7) into (2.1) yields the market-clearing price after  $\theta_{ipt+1}$  is realized:

$$p_{ipt+1} = \frac{\eta}{\eta - 1} \frac{w_{t+1}}{z_i} \frac{\theta_{ipt+1}^{(\eta-1)/\eta}}{\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt})}. \quad (2.9)$$

Thus, the expected price is a constant markup over marginal cost,  $\mathbb{E}_{\theta_{ipt+1}} [p_{ipt+1}] =$

$\frac{\eta}{\eta-1} \frac{w_{t+1}}{z_i}$ . Combining this with (2.7) gives optimal expected per-period profit:

$$\bar{\pi}_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) = \frac{(\eta-1)^{\eta-1}}{\eta^\eta} \left( \frac{z_i}{w_{t+1}} \right)^{\eta-1} P_{pt+1}^\eta C_{pt+1} \left[ \mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \right]^\eta - w_{t+1} f^o. \quad (2.10)$$

The quantity and price equations motivate the empirical specification for estimating firm-product expertise in (3.40).<sup>13</sup>

### 2.3.4 Product Switching and Exit

Each firm  $i$  is endowed with  $n_i$  product lines, indexed by  $k \in \{1, \dots, n_i\}$ , representing capacity slots to which products can be assigned over time. Let  $p_{kt} \in \{1, \dots, N, \emptyset\}$  denote the product assigned to line  $k$  in period  $t$ , with  $p_{kt} = \emptyset$  for a vacant line. We write  $p_k$  when the time index is clear. Product assignments evolve endogenously through firms' product-switching decisions. When a firm searches, a destination product is randomly drawn for the line. With probability  $x$ , search succeeds and the destination replaces the existing product or fills a vacancy.<sup>14</sup> Product lines are therefore the units of capacity and decision-making, while product indices identify markets.

A product line's value depends on the firm's posterior beliefs about product-specific expertise. Given operating costs and a fixed search cost  $f^s$  in labor units per searching line, the firm faces two extensive-margin decisions per line: whether to retain or drop its product and whether to search for another. The firm may retain a product while searching for its replacement. If all active products are dropped, the firm exits

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<sup>13</sup>The model counterpart of real shipments is  $p_{ipt}y_{ipt}/P_{pt}$ . Combining (2.7) and (2.9) gives

$$\begin{aligned} \log \left( \frac{p_{ipt}y_{ipt}}{P_{pt}} \right) &= (\eta-1) \log \left( \frac{\eta-1}{\eta} \right) + (\eta-1) \log P_{pt} + \log C_{pt} \\ &\quad + (\eta-1) \log \left( \frac{z_i}{w_t} \right) + (\eta-1) \log \mathcal{B}_{ip,t-1} + \frac{\eta-1}{\eta} (\log \theta_{ip} + \varepsilon_{ipt}), \end{aligned} \quad (2.11)$$

where  $\mathcal{B}_{ip,t-1} \equiv \mathcal{B}_{i,t-1}(\hat{\theta}_{ip,t-1}, \tau_{ip,t-1})$ . Product-year effects absorb the aggregate terms, while firm and firm-product controls account for other sources of shipment variation. Appendix B.2 discusses the mapping from estimated fixed effects to structural expertise.

<sup>14</sup>The success probability  $x$  is analogous to R&D productivity in endogenous growth models. Firms choose search participation rather than intensity; the fixed cost  $f^s$  and  $x$  jointly determine participation and expected product additions.

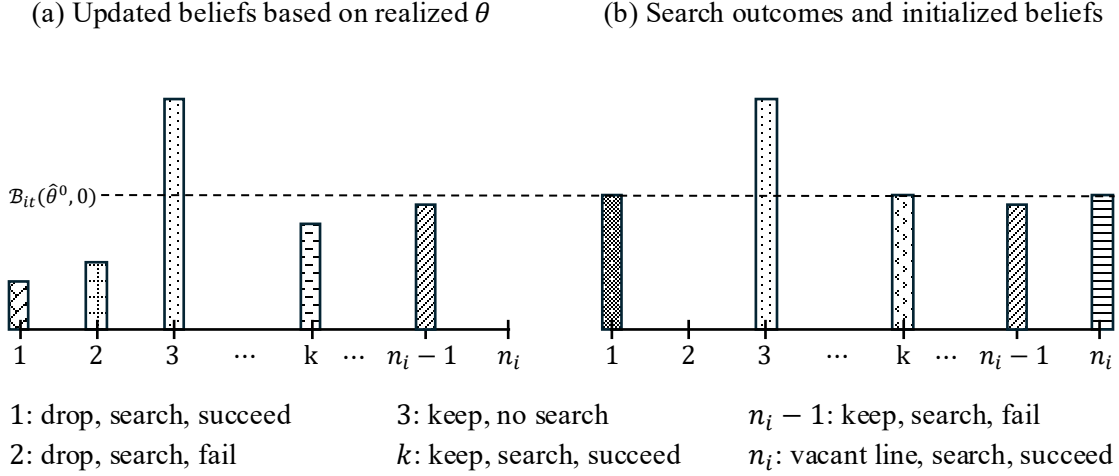


Figure 2: Evolution of product lines for firm  $i$  with potential product scope  $n_i = n(\varphi_i)$

before search; otherwise, it searches to fill newly and previously vacant lines. Firms exit exogenously with probability  $\delta$ . We characterize search and drop thresholds below.

Figure 2 illustrates these decisions and outcomes for firm  $i$  with  $n_i = n(\varphi_i)$ . Bar heights represent  $B_{it}(\hat{\theta}, \tau)$  after belief updating but before switching in panel (a), and after search outcomes in panel (b). Different patterns identify distinct products, while vacant lines have no bar. The dashed line indicates  $B_{it}(\hat{\theta}^0, 0)$  under the common prior.

Products in lines 1 and 2 lie below their drop thresholds and are dropped. The product in line 3 lies above its search threshold and is retained without search. Products in lines  $k$  and  $n_i - 1$  lie between their respective drop and search thresholds and are retained while searches are conducted. Because active products remain, the firm also searches to fill newly vacant lines 1 and 2 and the initially vacant line  $n_i$ .

Searches succeed for lines 1,  $k$ , and  $n_i$ , which receive new products with beliefs reset to the common prior. Searches fail for lines 2 and  $n_i - 1$ : line 2 remains vacant, while line  $n_i - 1$  retains its existing product. An initially vacant line would also remain vacant after failed search, a case not shown in the figure.

### 2.3.5 Value Function of Product Lines

Since product-switching decisions are made within product lines, we formulate the recursive problem at the product-line level. Endogenous exit, however, can link a line's continuation value to posterior beliefs about the firm's other products through the exit probability. We establish the conditions for portfolio independence in Section 2.3.7. We consider firms aged two or older; Section 2.4 presents the age-one startup problem.

Let  $V_{it}^{(\cdot)}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) \equiv V^{(\cdot)}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i, m_i)$ . We likewise suppress firm-level states in beliefs, thresholds, and transition probabilities when there is no ambiguity. For an occupied product line  $k$ , the value function is

$$V_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) = \max\{V_{it}^{ns}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}), V_{it}^{cs}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}), V_{it}^{ds}(\emptyset)\}, \quad (2.12)$$

where  $V_{it}^{ns}$  is the value of retaining the product without search,  $V_{it}^{cs}$  is the value of retaining it while searching, and  $V_{it}^{ds}(\emptyset)$  is the value of dropping it and searching, conditional on firm continuation. If all active products are dropped, the firm exits before search; the associated exit payoff is specified below. The corresponding search and drop thresholds,  $\mathcal{B}_{it}^{cs}(\tau_{ip_k t})$  and  $\mathcal{B}_{it}^{ds}(\tau_{ip_k t})$ , are characterized in Proposition 1.<sup>15</sup>

**(a) No search ( $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) \geq \mathcal{B}_{it}^{cs}(\tau_{ip_k t})$ ):** The firm retains the product without searching this period but may search in subsequent periods as beliefs evolve. The corresponding value is

$$V_{it}^{ns}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) = \beta \left[ \bar{\pi}_{it+1}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) + (1 - \delta) \mathbb{E}_{ip_k t} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) + \delta V_{it+1}^{\delta} \right], \quad (2.13)$$

where  $\mathbb{E}_{ip_k t}$  conditions on the current product-line and firm-level states, and  $V_{it+1}^{\delta}$  is the terminal value upon exogenous firm exit, which we normalize to zero. The expected

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<sup>15</sup>We focus on the case in which continuing firms search in vacant lines and  $V_{it}^{ds}(\emptyset) > 0$ , with the search value defined in (2.15). The condition for  $\mathcal{B}_{it}^{cs}(\tau_{ip_k t}) > \mathcal{B}_{it}^{ds}(\tau_{ip_k t})$  is given in (2.18).

continuation value incorporates next period's optimal choices in (2.12):

$$\mathbb{E}_{ip_k t} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) = \int_{\mathbb{R}} V_{it+1}(\hat{\theta}', \tau') g_{it}(\hat{\theta}', \tau' \mid \hat{\theta}_{ip_k t}, \tau_{ip_k t}) d\hat{\theta}',$$

where  $\tau' = \tau_{ip_k t} + 1$  and  $g_{it}$  is defined in (2.5). The corresponding decision regions are

$$\begin{aligned} \mathcal{R}_{ip_k t+1}^{ns} &= \{(\hat{\theta}', \tau') : \mathcal{B}_{it+1}(\hat{\theta}', \tau') \geq \mathcal{B}_{it+1}^{cs}(\tau')\}, \\ \mathcal{R}_{ip_k t+1}^{cs} &= \{(\hat{\theta}', \tau') : \mathcal{B}_{it+1}^{ds}(\tau') \leq \mathcal{B}_{it+1}(\hat{\theta}', \tau') < \mathcal{B}_{it+1}^{cs}(\tau')\}, \\ \mathcal{R}_{ip_k t+1}^{ds} &= \{(\hat{\theta}', \tau') : \mathcal{B}_{it+1}(\hat{\theta}', \tau') < \mathcal{B}_{it+1}^{ds}(\tau')\}. \end{aligned}$$

The last region corresponds to dropping the product and includes endogenous exit when all active products are dropped.

**(b) Continue/Switch with search ( $\mathcal{B}_{it}^{ds}(\tau_{ip_k t}) \leq \mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) < \mathcal{B}_{it}^{cs}(\tau_{ip_k t})$ ):** In this region, the firm searches for a new product while retaining the current product. If search succeeds, the new product replaces it at the end of the period; otherwise, the firm retains it. Using the no-search value in (2.13), we write

$$V_{it}^{cs}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) = -w_t f^s + x V_{it}^{ns}(\hat{\theta}_{ip'_k t}^0, 0) + (1 - x) V_{it}^{ns}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}), \quad (2.14)$$

where  $p'_k$  denotes the new destination product. Upon success, we treat the firm as having no prior experience in the destination market and reset its beliefs to the common prior, with  $\hat{\theta}_{ip'_k t}^0 = \theta_0$  and tenure zero.<sup>16</sup> Following failed search, the payoff coincides with the no-search value because the firm retains the existing product.

Search operates as follows. The firm's exclusion set contains its currently operated products and the most recent destination associated with each vacant line, yielding  $n(\varphi_i)$  distinct excluded products.<sup>17</sup> Destinations are drawn uniformly without replace-

<sup>16</sup>For tractability, we apply the prior-reset convention whenever a destination product is adopted. If the firm returns to a previously operated product, earlier information could be retained; we abstract from this history dependence. Our calibration uses a large product space,  $N = 1400$ .

<sup>17</sup>We assume  $n(\varphi_i) \leq \lfloor N/2 \rfloor$ , ensuring enough distinct destinations for all product lines.

ment across searching lines from the remaining  $N - n(\varphi_i)$  products. Search succeeds independently across lines with probability  $x$ , in which case the destination is adopted. If search on a vacant line fails, the unsuccessful draw becomes its most recent destination for the next search. Under uniform destination draws and symmetric product markets, expected continuation values are identical across feasible destinations, allowing us to suppress expectations over destination products.<sup>18</sup>

**(c) Drop/Switch with search ( $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) < \mathcal{B}_{it}^{ds}(\tau_{ip_k t})$ ):** In this case, the firm drops the current product and, conditional on firm continuation, searches to fill the vacant line. If search succeeds, the line is filled with the new product; otherwise, it remains vacant. Using (2.13), the value is

$$V_{it}^{ds}(\emptyset) = -w_t f^s + x V_{it}^{ns}(\hat{\theta}_{ip_k' t}, 0) + (1 - x)\beta [(1 - \delta)V_{it+1}^{ds}(\emptyset) + \delta V_{it+1}^{\delta}]. \quad (2.15)$$

After failed search, the focal line remains vacant. We assume that its payoff upon endogenous firm exit equals its value under firm continuation,  $V_{it+1}^{ds}(\emptyset)$ . Thus, the endogenous-exit probability drops out of the continuation value. Section 2.3.7 formalizes this payoff-equivalence assumption.

### 2.3.6 Search and Product-Drop Thresholds

Given the product-line value functions, we characterize the search and drop thresholds,  $\mathcal{B}_{it}^{cs}(\tau_{ip_k t})$  and  $\mathcal{B}_{it}^{ds}(\tau_{ip_k t})$ . We take stationary aggregate prices and quantities as given and suppress dependence on firm-level states when there is no ambiguity.

At the search threshold,  $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}^{cs}, \tau_{ip_k t}) = \mathcal{B}_{it}^{cs}(\tau_{ip_k t})$ , the firm is indifferent between retaining the product with and without search. Using (2.14), this gives

$$V_{it}^{ns}(\hat{\theta}_{ip_k' t}^0, 0) - V_{it}^{ns}(\hat{\theta}_{ip_k t}^{cs}, \tau_{ip_k t}) = \frac{w_t f^s}{x}. \quad (2.16)$$

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<sup>18</sup>Appendix A.3 shows that, despite sampling without replacement across searching lines, each line is marginally assigned any feasible destination with probability  $1/[N - n(\varphi_i)]$ .

Thus, at the search margin, the value gain from successfully switching to a new product equals the search cost scaled by the success probability.

Similarly, at the drop threshold,  $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}^{ds}, \tau_{ip_k t}) = \mathcal{B}_{it}^{ds}(\tau_{ip_k t})$ , the firm is indifferent between retaining the product while searching and dropping it:

$$V_{it}^{cs}(\hat{\theta}_{ip_k t}^{ds}, \tau_{ip_k t}) = V_{it}^{ds}(\emptyset). \quad (2.17)$$

Appendix A.2 derives the equivalent condition in more detail.

Holding tenure and firm-level states fixed, a higher posterior mean raises expected profits and shifts future beliefs upward without changing their conditional variance. The resulting monotonicity and single-crossing properties ensure that each indifference condition has at most one solution. Appendix A.2 establishes these properties and provides the supporting conditions for the following proposition. Fix product tenure, firm-level states, and aggregate prices and quantities. Suppose  $0 < x < 1$  and  $V_{it}^{ns}(\hat{\theta}, \tau) < V_{it}^{cs}(\hat{\theta}, \tau) < V_{it}^{ds}(\emptyset)$  for all sufficiently low  $\hat{\theta}$ . Under these conditions, the following holds:

**Proposition 1** (Existence, Uniqueness, and Ordering of Thresholds). *There exist unique finite cutoffs  $\hat{\theta}^{ds}$  and  $\hat{\theta}^{cs}$  satisfying (2.17) and (2.16), respectively. Moreover,*

$$\hat{\theta}^{cs} > \hat{\theta}^{ds} \iff V_{it}^{ns}(\hat{\theta}^0, 0) - V_{it}^{ns}(\hat{\theta}^{ds}, \tau) > \frac{w_t f^s}{x}. \quad (2.18)$$

When (2.18) holds, the optimal policy is to drop the product if  $\hat{\theta} < \hat{\theta}^{ds}$ , retain it while searching if  $\hat{\theta}^{ds} \leq \hat{\theta} < \hat{\theta}^{cs}$ , and retain it without search if  $\hat{\theta} \geq \hat{\theta}^{cs}$ . After dropping, the firm searches to fill the vacant line conditional on firm continuation.

Since  $\mathcal{B}_{it}(\hat{\theta}, \tau)$  is strictly increasing in  $\hat{\theta}$ , these cutoffs uniquely define  $\mathcal{B}_{it}^{ds}(\tau)$  and  $\mathcal{B}_{it}^{cs}(\tau)$ , with  $\mathcal{B}_{it}^{ds}(\tau) < \mathcal{B}_{it}^{cs}(\tau)$  under (2.18). Restoring the suppressed arguments, both thresholds depend on  $(\tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i, m_i)$ .

*Proof.* See Appendix A.2. □

Condition (2.18) requires the value gain from successfully switching to a new product at the drop threshold to exceed the search cost scaled by the success probability. It

ensures a nonempty region in which the firm retains its current product while searching for a replacement. Holding the value difference fixed, a lower search cost or higher success probability favors this intermediate region.

Product tenure affects the thresholds separately from firm age. Holding the posterior mean and firm-level states fixed, longer tenure changes posterior variance and therefore both  $\mathcal{B}_{it}$  and the distribution of future beliefs. Thus, firms with the same age and information-processing ability can face different thresholds depending on their experience with a particular product.

### 2.3.7 Firm Value Function

Let  $\Theta_{it} \equiv \{(\hat{\theta}_{ip_{kt}}, \tau_{ip_{kt}})\}_{k=1}^{n_i}$  denote firm  $i$ 's portfolio of product-line states, with  $\emptyset$  denoting a vacant line. The firm exits after product-drop decisions and before search if all active products are dropped. Thus, whether a dropped or vacant line reaches search can depend on the states and drop decisions of the firm's other product lines.

We remove portfolio dependence by assuming each line's payoff upon endogenous firm exit equals the value of a vacant line under firm continuation,  $V_{it}^{ds}(\emptyset)$ .<sup>19</sup> The following establishes product-line separability under this payoff-equivalence assumption.

**Lemma 2** (Product-Line Separability). *Under the payoff-equivalence assumption, conditional on  $(a_{it}, \psi_{it}, \varphi_i, m_i)$ , the value of product line  $k$  depends only on its own state  $(\hat{\theta}_{ip_{kt}}, \tau_{ip_{kt}})$  and is independent of posterior beliefs in the firm's other product lines.*

*Proof.* See Appendix A.3. □

Operating profits and operating and search fixed costs are additive across product lines, and we exclude firm-level fixed operating costs. Together with product-line separability, this implies that firm value equals the sum of its product-line values.

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<sup>19</sup>This assumption does not imply that exiting firms search; endogenous exit occurs before search. The payoff upon exogenous exit remains normalized to zero,  $V^\delta = 0$ .

**Proposition 2** (Additive Firm Value). *The firm-level value function is additively separable across product lines:*

$$V(\Theta_{it}; a_{it}, \psi_{it}, \varphi_i, m_i) = \sum_{k=1}^{n_i} V(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i, m_i),$$

where a vacant line contributes  $V^{ds}(\emptyset; a_{it}, \psi_{it}, \varphi_i, m_i)$ .

*Proof.* Follows from Lemma 2 and additivity of product-line payoffs.  $\square$

## 2.4 Potential Startups

Each period, potential startups make entry and capability-investment decisions concurrently with incumbents' search decisions. Pre-entry investments  $(a_t^m, a_t^z)$  determine managerial and productivity capabilities,  $m_{0t}$  and  $z_{0t}$ , according to

$$\log m_{0t} = (1 - \zeta_m) \log a_t^m + \zeta_m \log M_{t-1}, \quad \log z_{0t} = (1 - \zeta_z) \log a_t^z + \zeta_z \log Z_{t-1}. \quad (2.19)$$

Here,  $M_{t-1}$  and  $Z_{t-1}$  denote aggregate managerial and productivity capabilities inherited from the previous cohort, with  $M_{t-1} = m_{0,t-1}$  and  $Z_{t-1} = z_{0,t-1}$  in equilibrium.  $\zeta_m, \zeta_z \in [0, 1)$  govern inter-cohort transferability.<sup>20</sup>

The startup resource cost, in labor units, is  $f_t^e(a_t^m, a_t^z) = \bar{f}^e + \alpha_m^e (a_t^m)^\chi + \alpha_z^e (a_t^z)^\chi$ , where  $\bar{f}^e$  is the fixed startup cost,  $\alpha_m^e$  and  $\alpha_z^e$  scale investment costs, and  $\chi$  governs their curvature. Potential startups take these parameters and inherited capabilities as given. For the entering cohort,  $m_{0t}$  shifts the distribution of managerial ability  $m_i$ , while  $z_{0t}$  determines productivity through  $z_i = z_{0t} \varphi_i$ .

Upon entry, startups draw permanent fundamental types  $(\varphi_i, m_i)$ , with  $\varphi_i \sim H_\varphi$  and  $m_i \mid \varphi_i \sim H_m(\cdot \mid \varphi_i; m_{0t})$ . The firm observes its productive potential  $\varphi_i$  during pro-

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<sup>20</sup>In the spirit of “standing on the shoulders of giants,” this specification combines current investment with inherited knowledge (Romer, 1990; Jones, 1995). Evidence on entrepreneurial and managerial knowledge transmission motivates  $\zeta_m > 0$  (Guiso et al., 2021; Bloom et al., 2019; Giordelli, 2019), while models of technology and idea diffusion motivate  $\zeta_z > 0$  (Perla and Tonetti, 2014; Sampson, 2016; Buera and Oberfield, 2020).

duction planning and decides whether to operate.<sup>21</sup> Entry is undirected: each startup receives one initial product market, with its remaining  $n(\varphi_i) - 1$  product lines vacant.<sup>22</sup> Conditional on operating, the firm produces and sells its initial product in  $t + 1$ . Managerial ability  $m_i$ , unknown during entry and initial production, is revealed before the first Bayesian update about product-specific expertise.

After the first update, the age-one startup chooses whether to retain its initial product and whether to search for its replacement. Retention also allows the firm to search in its  $n(\varphi_i) - 1$  vacant lines; dropping the initial product triggers exit before search.<sup>23</sup> Let  $V_{t+1}^{e,1}(\hat{\theta}; \psi, \varphi, m)$  denote firm value at this decision, when firm age and initial-product tenure equal one. Suppressing common states  $(1, \psi, \varphi, m)$  on the right-hand side,

$$V_{t+1}^{e,1}(\hat{\theta}; \psi, \varphi, m) = \max \left\{ V_{t+1}^{ns}(\hat{\theta}, 1) + (n(\varphi) - 1)V_{t+1}^{ds}(\emptyset), \right. \\ \left. V_{t+1}^{cs}(\hat{\theta}, 1) + (n(\varphi) - 1)V_{t+1}^{ds}(\emptyset), V_{t+1}^{ds}(\emptyset) \right\}. \quad (2.20)$$

Thus, the initial retention decision accounts for the search opportunities preserved in the other lines.

Let  $\tilde{V}_t^e(\varphi)$  denote operating value after  $\varphi$  is observed but before  $m$  is revealed. Then

$$\tilde{V}_t^e(\varphi) = \beta \bar{\pi}_{it+1}(\hat{\theta}_{ip_1t}^0, 0; 0, \varphi) + \beta(1 - \delta) \mathbb{E}_{m|\varphi} \left[ \mathbb{E}_{\hat{\theta}' | (\hat{\theta}_{ip_1t}^0, 0), m} V_{t+1}^{e,1}(\hat{\theta}'; \psi_{it+1}, \varphi, m) \right]. \quad (2.21)$$

The startup's operating option and operating cutoff  $\underline{\varphi}_t^e$  satisfy

$$V_t^e(\varphi) = \max\{\tilde{V}_t^e(\varphi), 0\}, \quad (2.22)$$

$$\tilde{V}_t^e(\underline{\varphi}_t^e) = 0. \quad (2.23)$$

<sup>21</sup>Pre-entry investments precede the realization of  $\varphi_i$ . Prototyping or production planning reveals productive potential, allowing startups to abandon before production after an unfavorable realization.

<sup>22</sup>Uniform assignments and symmetric product markets allow us to suppress expectations over initial product assignments.

<sup>23</sup>Upon endogenous exit at age one, the initial line receives  $V_{t+1}^{ds}(\emptyset)$ , while initially vacant lines receive zero. From age two onward, the payoff-equivalence assumption in Section 2.3.7 applies to all lines.

Before entry, potential startups choose  $(a_t^m, a_t^z)$  to maximize expected net entry value:

$$(a_t^m, a_t^z) \in \arg \max_{a^m, a^z > 0} \{ \mathbb{E}_\varphi[V_t^e(\varphi)] - w_t f_t^e(a^m, a^z) \}. \quad (2.24)$$

We suppress the resulting dependence of  $V_t^e(\varphi)$  on  $(m_{0t}, z_{0t})$ . Free entry implies

$$\mathbb{E}_\varphi[V_t^e(\varphi)] = w_t f^e, \quad (2.25)$$

where  $f^e \equiv f_t^e(a_t^m, a_t^z)$  is the startup resource cost at the optimum. In stationary equilibrium,  $a^m = m_0$  and  $a^z = z_0$ . The mass of entrants is determined in equilibrium.

## 2.5 Equilibrium

### 2.5.1 Labor Market Clearing

Let  $\xi_{ipt} \equiv (\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i, m_i) \in \mathcal{X}$  denote the state of an active product line after period- $t$  search and entry, where  $\mathcal{X}$  is the active product-line state space. From (2.7), its variable production labor demand in  $t + 1$  is

$$\ell_{ipt+1}(\xi_{ipt}) = \left( \frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} [\mathcal{B}(\xi_{ipt})]^\eta z(\varphi_i; z_0)^{\eta-1} w_{t+1}^{-\eta}. \quad (2.26)$$

Let  $\mu_{pt}$  denote the measure of active product lines over  $\mathcal{X}$  in market  $p$  at the end of period  $t$ . Aggregating across these lines gives total production labor demand:

$$L_{pt+1} = \left( \frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} w_{t+1}^{-\eta} \int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi; z_0)^{\eta-1} \mu_{pt}(d\xi). \quad (2.27)$$

The labor market clearing condition is

$$\bar{L} = \sum_{p=1}^N L_{pt+1} + \mathcal{F}_{t+1}^o + \mathcal{F}_{t+1}^s + \mathcal{F}_{t+1}^e, \quad (2.28)$$

where  $\mathcal{F}_{t+1}^o$ ,  $\mathcal{F}_{t+1}^s$ , and  $\mathcal{F}_{t+1}^e$  denote labor used for operating fixed costs, search costs, and startup costs (including capability investments), respectively. Section 2.5.3 defines these aggregates.

### 2.5.2 Firm and Product-Line Distributions

Let  $\xi_{it}^f \equiv (\Theta_{it}, a_{it}, \psi_{it}, \varphi_i, m_i) \in \mathcal{X}^f$  denote the firm-level state.<sup>24</sup> Let  $\Gamma_t$  denote the cross-sectional measure of firms over  $\mathcal{X}^f$  after period- $t$  search and entry.

Given equilibrium prices and policies,  $\mathcal{Q}_{t+1}^f(A \mid \xi^f)$  is the probability that a firm in state  $\xi^f$  survives to the end of period  $t + 1$  with its state in  $A \subseteq \mathcal{X}^f$ . This transition incorporates Bayesian updating, product-drop and search decisions, search outcomes and destination assignments, and changes in age and information-processing ability. The remaining probability corresponds to exogenous death or endogenous exit.

The distribution of startup states conditional on operating is

$$\Gamma_t^e(A) \equiv \frac{1}{N[1 - H_\varphi(\underline{\varphi}_t^e)]} \sum_{p=1}^N \int_{\varphi \geq \underline{\varphi}_t^e} \int \mathbf{1}_{\{\xi_t^{f,e}(p, \varphi, m) \in A\}} H_m(dm \mid \varphi; m_{0t}) H_\varphi(d\varphi), \quad (2.29)$$

where  $\xi_t^{f,e}(p, \varphi, m)$  denotes an age-zero firm with one product assigned to market  $p$  at the common prior and  $n(\varphi) - 1$  vacant lines. The firm measure evolves according to

$$\Gamma_{t+1}(A) = \int_{\mathcal{X}^f} \mathcal{Q}_{t+1}^f(A \mid \xi^f) \Gamma_t(d\xi^f) + \mathcal{M}_{t+1}^e \Gamma_{t+1}^e(A), \quad (2.30)$$

where  $\mathcal{M}_{t+1}^e$  is the mass of operating entrants.

The active product-line measure  $\mu_{pt}$  used in market clearing is obtained by projecting  $\Gamma_t$  onto active lines in market  $p$ . For  $A \subseteq \mathcal{X}$ ,

$$\mu_{pt}(A) = \int_{\mathcal{X}^f} \sum_{k=1}^{n(\varphi)} \mathbf{1}_{\{k \text{ active}, p_k=p\}} \mathbf{1}_{\{\xi_k \in A\}} \Gamma_t(d\xi^f), \quad (2.31)$$

---

<sup>24</sup> $\Theta_{it}$  also records product assignments and, for the search exclusion set, the most recent destination associated with each vacant line. Initially vacant startup lines have no previous destination.

where  $p_k$  and  $\xi_k$  denote line  $k$ 's product assignment and state within  $\xi^f$ . Thus,  $\mu_{pt}$  captures the mass and state distribution of active product lines in market  $p$ .

In stationary equilibrium, the firm measure satisfies

$$\Gamma(A) = \int_{\mathcal{X}^f} \mathcal{Q}^f(A \mid \xi^f) \Gamma(d\xi^f) + \mathcal{M}^e \Gamma^e(A), \quad (2.32)$$

with  $\mu_p$  given by (2.31).<sup>25</sup>

### 2.5.3 Aggregate Variables

Since each active product line uses  $f^o$  units of labor for its operating fixed cost, total labor used for operating fixed costs in period  $t$  is

$$\mathcal{F}_t^o = f^o \sum_{p=1}^N \mu_{p,t-1}(\mathcal{X}). \quad (2.33)$$

Let  $\mathcal{S}_t(\xi^f)$  denote the expected number of product-line searches by a firm starting period  $t$  in state  $\xi^f$ , accounting for within-period uncertainty, exogenous death, and endogenous exit before search. Total labor used for search is

$$\mathcal{F}_t^s = f^s \int_{\mathcal{X}^f} \mathcal{S}_t(\xi^f) \Gamma_{t-1}(d\xi^f). \quad (2.34)$$

Online Appendix OA.A.6 provides the expression using product-line measures.

Because only a fraction  $1 - H_\varphi(\underline{\varphi}_t^e)$  of startups operate, the gross mass paying pre-entry costs is  $\mathcal{M}_t^e / [1 - H_\varphi(\underline{\varphi}_t^e)]$ . Total labor used for startup costs is therefore

$$\mathcal{F}_t^e = f_t^e(a_t^m, a_t^z) \frac{\mathcal{M}_t^e}{1 - H_\varphi(\underline{\varphi}_t^e)}. \quad (2.35)$$

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<sup>25</sup>Tracking portfolio states becomes computationally infeasible as scope increases. We compute equilibrium using product-line measures, including vacancies, and approximate endogenous exit probabilities from these marginals. Online Appendix OA.A.5 provides the laws of motion and approximation.

Substituting the market-clearing price in (2.9) into the product-level price index gives

$$P_{pt} = \frac{\eta}{\eta - 1} w_t \left[ \int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi; z_0)^{\eta-1} \mu_{p,t-1}(d\xi) \right]^{-\frac{1}{\eta-1}}. \quad (2.36)$$

Appendix A.4 provides the derivation.

In a symmetric equilibrium, production labor demand is identical across product markets, so (2.28) implies  $L_{pt} = (\bar{L} - \mathcal{F}_t^o - \mathcal{F}_t^s - \mathcal{F}_t^e)/N$ . Combining this condition with (2.27) and (2.36) gives

$$C_{pt} = \frac{\bar{L} - \mathcal{F}_t^o - \mathcal{F}_t^s - \mathcal{F}_t^e}{N} \left[ \int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi; z_0)^{\eta-1} \mu_{p,t-1}(d\xi) \right]^{\frac{1}{\eta-1}}. \quad (2.37)$$

Aggregate consumption is then  $\bar{C}_t = N^{\nu/(\nu-1)} C_{pt}$ .

#### 2.5.4 Symmetric Stationary Recursive Competitive Equilibrium

**Definition 1.** A symmetric stationary recursive competitive equilibrium is a collection

$$\left\{ C, P, w, \mathcal{M}^e, a^m, a^z, m_0, z_0, y, p, \ell, L, \mathcal{F}^o, \mathcal{F}^s, \mathcal{F}^e, \right. \\ \left. V, V^{ns}, V^{cs}, V^{ds}, V^{e,1}, \tilde{V}^e, V^e, \mathcal{B}^{cs}, \mathcal{B}^{ds}, \underline{\varphi}^e, \Gamma, \Gamma^e, \mu \right\}$$

such that: (i) posterior beliefs follow (2.3) and (2.4), with  $\mathcal{B}$  given by (2.8); (ii) given  $C$ ,  $P$ ,  $w$ , and beliefs, output  $y$ , prices  $p$ , and variable production labor demand  $\ell$  satisfy (2.7), (2.9), and (2.26), respectively; (iii) product-line values satisfy (2.12), (2.13), (2.14), and (2.15), and the search and drop thresholds satisfy (2.16) and (2.17); (iv) startup values, the operating cutoff, and capability investments satisfy (2.20), (2.21), (2.22), (2.23), and (2.24), with free entry given by (2.25); (v) capabilities satisfy (2.19), with  $M = m_0$ ,  $Z = z_0$ , and hence  $a^m = m_0$  and  $a^z = z_0$ ; (vi)  $\Gamma^e$  is given by (2.29),  $\Gamma$  is invariant under (2.32), and  $\mu$  is its product-line projection in (2.31); (vii)  $L$ ,  $\mathcal{F}^o$ ,  $\mathcal{F}^s$ , and  $\mathcal{F}^e$  satisfy (2.27), (2.33), (2.34), and (2.35), and the labor market clears according to (2.28); and (viii)  $P$  and  $C$  satisfy (2.36) and (2.37), with symmetry across product markets requiring

$P_p = P$ ,  $C_p = C$ ,  $L_p = L$ , and  $\mu_p = \mu$  for all  $p$ .

Because only relative prices matter, we normalize the stationary wage to one,  $w = 1$ .

## 2.6 Model Implications

### 2.6.1 Output Wedge under Imperfect Learning

Building on Lemma 1, we characterize production distortions relative to the benchmark in which  $\theta_{ip}$  is known, holding aggregate prices and quantities fixed. Firms with lower information-processing ability, particularly when they are young, place greater weight on the prior. Together with posterior uncertainty, this generates expected underproduction of sufficiently high-expertise products and overproduction of sufficiently low-expertise products before endogenous selection. Since households place greater weight on high-expertise products, these distortions provide a potential welfare-loss channel. The following proposition formalizes this distortion.

**Proposition 3** (Learning Frictions and Output Wedges). *Let  $y_{ipt}^{FI}$  denote optimal output when  $\theta_{ip}$  is known but the contemporaneous demand shock remains unobserved. Define the output wedge by*

$$1 - \omega_{ipt} \equiv \frac{y_{ipt}^*}{y_{ipt}^{FI}} = \frac{\left[ \mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta}{\theta_{ip}^{\eta-1} \exp \left\{ \frac{(\eta-1)^2}{2\eta} \sigma_\varepsilon^2 \right\}}. \quad (2.38)$$

For  $\hat{\sigma}_{ipt-1}^2 > 0$ , conditional on true  $\theta_{ip}$ , the expected wedge is positive if and only if

$$\log \theta_{ip} > \theta_0 + \frac{\eta-1}{2} \left[ \left( 1 + \frac{1}{\eta} \right) \sigma_0^2 - \hat{\sigma}_{ipt-1}^2 \right], \quad (2.39)$$

and negative when the inequality is reversed.<sup>26</sup> At a given product tenure, lower information-processing ability raises posterior variance and therefore lowers this cutoff, expanding the range of expertise values associated with underproduction in expectation.

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<sup>26</sup>Expectations average over potential signal histories at fixed tenure and information-processing path, without conditioning on endogenous product retention.

*Proof.* See Appendix A.5. □

These comparisons isolate the intensive-margin effects of learning. Average wedges among continuing products additionally reflect endogenous selection.

### 2.6.2 Product Switching under Imperfect Learning

Firms retain uncertain products partly because further experience may show that they are profitable. This option value reflects their ability to adjust future production and product-switching decisions as they learn, as formalized in Lemma A.1 in Appendix A.6. Greater information-processing ability changes this option value in two ways: more is already known about existing products, while new products can be evaluated more quickly. We examine how these forces affect the drop and search thresholds.

Comparisons in  $\psi_{it}$  encompass managerial differences at a given firm age and age differences at a given managerial ability, including the information histories and future paths implied by the model. Product tenure, productivity, and aggregate prices and quantities are held fixed; we then consider productivity and tenure separately.

**Proposition 4** (Information Processing and Product Switching). *Suppose the continue-and-search region is nonempty at all relevant firm ages.*

- (i) *Information processing. Higher  $\psi_{it}$  raises both thresholds if it reduces the incumbent's remaining learning value and weakly improves replacement opportunities at each date, with a strict current learning effect.*<sup>27</sup>
- (ii) *Productivity. For a given information-processing path, higher  $z_i$  lowers the drop threshold if retention has a larger expected discounted fixed-cost burden than a vacant line from next period. It raises the search threshold if the incumbent's excess fixed-cost burden relative to a fresh product is less than  $f^s/x$ .*

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<sup>27</sup>Replacement opportunities are measured by the value of a fresh product relative to a vacant line from next period, net of the search cost per successful replacement; see (A.7).

(iii) *Product tenure. At the same firm age and information-processing path, longer firm-product tenure raises both thresholds.*

*Proof.* See Appendix A.7. □

Under the stated conditions, information processing and productivity both encourage search but have opposite effects on retention: greater information-processing ability makes firms more selective, whereas higher productivity makes retaining weak products more profitable. The continue-and-search region widens with productivity under part (ii), and with information-processing ability under the additional sufficient condition in (A.10); the effect of tenure on its width is ambiguous. These decisions shape product portfolios and thereby affect firm continuation and growth, as quantified in Section 4.6. Observed switching also depends on the distribution of demand terms.

### 2.6.3 Returns to Pre-Entry Investment

At fixed aggregate prices and quantities, expected entry value before investment costs satisfies  $\partial \mathbb{E}_\varphi[V^e(\varphi)]/\partial m_0 \geq 0$  and  $\partial \mathbb{E}_\varphi[V^e(\varphi)]/\partial z_0 > 0$ .<sup>28</sup> Higher  $m_0$  makes signals more informative, improving production and product-switching decisions. Since firms can disregard additional information, expected entry value cannot fall. Higher  $z_0$  raises operating profitability. Optimal pre-entry investments balance these marginal returns against acquisition costs.

Better information can increase the return to productivity by improving production and product-switching decisions. Complementarity is not automatic, however, because managerial improvements can raise value through fixed-cost savings. Appendix A.8 shows that the capabilities are locally complementary if and only if higher  $m_0$  weakly raises expected discounted gross operating payoffs.

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<sup>28</sup>The type distributions and regularity conditions are specified in Appendix A.8. The strict productivity sign requires production with positive probability.

### 3 Empirical Analysis

The model highlights how imperfect learning shapes product selection and firm growth. We turn to firm-product-level data from U.S. manufacturing to examine evidence for this mechanism. We estimate firms’ product-specific expertise and study how product-switching decisions vary with expertise, firm age, and information processing. We examine the relationship between product switching and firm growth and performance.

#### 3.1 Data and Measurement

Our main data source is the U.S. Census Bureau’s quinquennial Census of Manufactures (CMF), linked to the Longitudinal Business Database (LBD). Our sample covers eight census years from 1977 to 2012. The CMF covers U.S. manufacturing establishments with at least one paid employee and reports product codes—five-digit SIC or seven-digit NAICS codes—and the value of shipments by product. We link CMF establishments to the LBD and use firm identifiers to aggregate establishment-level observations to the firm level. Online Appendix [OA.D.1](#) describes the data sources and construction.

In addition, we use the Management and Organizational Practices Survey (MOPS), an establishment-level survey of management practices and organizational structure in U.S. manufacturing, with waves for 2010, 2015, and 2021. We use the average structured management score developed by [Bloom et al. \(2019\)](#), data availability (Q27), and the intensity of data use in decision making (Q28). We aggregate all three establishment-level measures to the firm level using value-of-shipments weights.

##### 3.1.1 Product Definition and Characteristics

We define products at the five-digit SIC level, using 1987 SIC codes throughout the sample. For 1977–1997, we use product codes that the Census Bureau concords to 1987 SIC codes. For subsequent census years, we construct concordances using the 1997 CMF and successive Census bridge files, as detailed in Online Appendix [OA.D.1](#).

The CMF reports the value of shipments by product and, for some products, physical quantities. We require positive shipment values and exclude administrative-record establishments, which need not report product-level shipments. We restrict the sample to observations in the Census Bureau’s “tabbed sample,” used for official tabulations. We deflate shipment values to 1997 dollars using NBER-CES industry-level shipment deflators (Becker et al., 2013). We measure firm-product tenure as the number of years the firm is observed producing the product, supplementing CMF with product-firm records from the Annual Survey of Manufactures (ASM).

We identify product additions and drops between consecutive census years. A firm adds a product between  $t$  and  $t + 5$  if it does not produce the product in  $t$  but produces it in  $t + 5$ . A firm drops a product if it produces the product in  $t$  but not in  $t + 5$ . We construct these indicators for census years  $t$  from 1982 to 2007.<sup>29</sup> We distinguish initial additions from re-entries and temporary drops from permanent drops. An initial addition is a product the firm has not previously been observed producing; a permanent drop is a product that does not reappear in the firm’s subsequent observed product portfolio. Our baseline analysis focuses on initial additions and permanent drops.

### 3.1.2 Firm Characteristics and Identifiers

We use firm age constructed by the Census Bureau following Haltiwanger et al. (2013). When a firm first appears, its age equals that of its oldest establishment; firm age increases annually thereafter. We define young firms as those aged five or less. We use the 1972 CMF to adjust left-censored ages and reduce misclassification of young firms. We measure firm size by total employment or total value of shipments. Our baseline analysis uses the Census Bureau’s firm identifier, `lbfid`; Online Appendix OA.D.1 describes the age adjustment and alternative longitudinal firm identifiers constructed following Dent et al. (2018). All nominal values are deflated to 1997 dollars using industry-level deflators for the relevant variables.

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<sup>29</sup>We use the 1977 CMF to help identify prior production histories and measure firm-product tenure. The 2012 CMF provides the endpoint for the final five-year interval.

Table 1: Prevalence of Product Switching and Multi-Product Firms

|                                                          | All firms |        | Young firms |        | Mature firms |        |
|----------------------------------------------------------|-----------|--------|-------------|--------|--------------|--------|
|                                                          | Firms     | Output | Firms       | Output | Firms        | Output |
| <i>Panel A: Product switching among continuing firms</i> |           |        |             |        |              |        |
| Any switching                                            | 0.646     | 0.910  | 0.667       | 0.694  | 0.642        | 0.913  |
| Add only                                                 | 0.128     | 0.067  | 0.130       | 0.139  | 0.128        | 0.066  |
| Drop only                                                | 0.182     | 0.169  | 0.143       | 0.146  | 0.187        | 0.169  |
| Add and drop                                             | 0.336     | 0.674  | 0.394       | 0.409  | 0.327        | 0.678  |
| <i>Panel B: Multi-product firms in the full sample</i>   |           |        |             |        |              |        |
| Multi-product firms                                      | 0.450     | 0.891  | 0.432       | 0.615  | 0.452        | 0.896  |

*Notes:* Entries are firm and output shares within each age group. Young firms are aged five or less; mature firms are aged six or more. Products are defined by five-digit SIC codes. Panel A covers continuing firms and measures additions and drops over five-year intervals; any switching is the sum of the three mutually exclusive categories. Panel B includes entrants and exiters. Online Appendix Tables [OA.12](#) and [OA.13](#) report the detailed results, including alternative samples and product definitions.

**Summary Statistics** Table [OA.11](#) in the Online Appendix reports means, standard deviations, and observation counts. The firm-level sample contains approximately 741,000 firm-year observations for firms with estimable product-specific expertise, requiring observations in at least two census years. Firms aged five or less account for 13% of these observations. Young firms are substantially smaller, employing 32.5 workers on average compared with 131.3 among mature firms. The table summarizes the firm-product characteristics and management measures used in the empirical analysis.

## 3.2 Empirical Results

Using these data, we examine how firms' product-switching decisions vary with product-specific expertise, firm age, and information processing, and how these decisions relate to firm growth and performance.

### 3.2.1 Prevalence of Product Switching

Panel A of Table 1 reports shares of firms engaging in product switching and their output shares by age group. To distinguish switching within continuing firms from

changes associated with entry and exit, we focus on continuing firms. In the pooled sample, 65% of continuing firms switch products over a five-year interval, accounting for 91% of output among continuing firms. These patterns underscore the prevalence and economic importance of product switching, consistent with [Bernard et al. \(2010\)](#).

Product switching often involves both adding and dropping products, which we refer to as product churning. Approximately 52% of product-switching firms (34% out of 65%) both add and drop products; the remainder either add or drop products. Product churning is more prevalent among young firms, whereas dropping products without adding others is less common than among mature firms.

Panel B reports the prevalence of multi-product firms under our baseline five-digit definition. Multi-product firms constitute 45% of firms but account for 89% of output. They are prevalent even among young firms, accounting for 43% of firms and 62% of output in this age group. Online Appendix [OA.D.3](#) reports detailed statistics, including diversification across industries and sectors. Young continuing firms produce fewer products than mature firms (1.6 versus 2.3), but add more (1.6 versus 1.2) and drop fewer (0.8 versus 1.0), as shown in Online Appendix Table [OA.14](#).

### **3.2.2 Product-Specific Expertise Estimation**

We estimate firms' product-specific expertise to examine its relationship with product selection and resource reallocation. The CMF reports the value of shipments at the firm-product level. Shipment values, however, reflect both expertise and other determinants of product performance, including product life cycle dynamics ([Argente et al., 2024](#)), customer capital accumulation, and transitory demand shocks.

Our empirical specification is motivated by equation (2.11), which relates log real shipments to product-specific expertise, firms' beliefs and productivity, and product-market conditions. We capture persistent differences in expertise through firm-product

fixed effects, in the spirit of [Bonhomme et al. \(2019\)](#), and estimate:

$$y_{ipjt} = \theta_{ip} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_{pt} + \varepsilon_{ipjt}, \quad (3.40)$$

where  $y_{ipjt}$  denotes log real shipments of product  $p$  by firm  $i$  in industry  $j$  and census year  $t$ . The firm-product fixed effect  $\theta_{ip}$  captures persistent differences in shipment values conditional on controls and fixed effects. We interpret this component as an empirical measure of product-specific expertise, reflecting how well a firm is suited to a product. Appendix [B.2](#) discusses its mapping to expertise in the model.

$X_{ipjt}$  includes firm-product tenure and its square to account for product life cycle dynamics within firms, including changes in shipments as firms gain production experience and build a customer base. Firm-level controls,  $X_{ijt}$ , account for differences in scale, resources, and production inputs that affect shipments across a firm's products.<sup>30</sup> Product scope and total assets also help account for alternative explanations for product-switching patterns over the firm life cycle, as discussed below. Product-year fixed effects,  $\delta_{pt}$ , absorb demand and technology conditions common to firms producing the same product in a year, including persistent differences across product markets.

We estimate  $\theta_{ip}$  using the 1977–2012 CMF, supplemented with ASM observations from 1976–2016. For ease of interpretation, we rescale the estimated fixed effects to the unit interval as  $q_{ip} \equiv (\hat{\theta}_{ip} - \underline{\theta})/(\bar{\theta} - \underline{\theta})$ , where  $\underline{\theta}$  and  $\bar{\theta}$  denote the minimum and maximum estimated fixed effects in the sample. This transformation preserves their ordering. Section [3.3](#) examines sensitivity to estimation windows, specifications, normalization methods, and statistical inference.

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<sup>30</sup>These controls include log employment, firm age, total assets, number of products, number of establishments, advertising and software expenditures, skill and capital intensity, and productivity. Skill and capital intensity are measured by the log ratios of nonproduction employment and capital to total employment, respectively. We also include the logs of production employment, equipment and structures capital stocks, material costs, and energy costs.

### 3.2.3 Product Selection

The model links product selection to firms' information about their product-specific expertise. Using our expertise measure, we examine which products firms retain, drop, and add, and how these decisions vary with firm age, management practices, and data availability, which shape information-processing precision in the model.

**Selection on Product Performance** We first examine whether firms are more likely to drop products with lower contemporaneous shipments or estimated expertise. This establishes the overall pattern of selection before examining how it varies with firm age and information processing. Let  $\mathcal{I}_{ipjt}^{drop}$  equal one if firm  $i$  in industry  $j$  permanently drops product  $p$  between census years  $t$  and  $t + 5$ . Our specifications are nested in:

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta_1 Q_{ipt} + \beta_2 \mathcal{I}_{ijt}^{young} + \beta_3 Q_{ipt} \times \mathcal{I}_{ijt}^{young} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}. \quad (3.41)$$

Here,  $Q_{ipt}$  denotes either log real shipments or normalized product-specific expertise,  $q_{ip}$ , and  $\mathcal{I}_{ijt}^{young}$  equals one if firm  $i$  is aged five or less in year  $t$ . The vectors  $X_{ipjt}$  and  $X_{ijt}$  contain the firm-product and firm-level controls used in expertise estimation, respectively. When  $Q_{ipt} = q_{ip}$ , we also control for log real shipments to distinguish selection on expertise from selection on contemporaneous performance. Firm fixed effects,  $\delta_i$ , absorb persistent differences across firms, while product-year fixed effects,  $\delta_{pt}$ , absorb conditions common to firms operating in the same product market.

The first two columns of Table 2 report estimates with all controls, omitting the young-firm indicator and its interaction with product performance.<sup>31</sup> Column 1 shows that firms are more likely to drop products with lower contemporaneous shipments. Column 2 shows that firms are also more likely to drop products with lower expertise, even conditional on contemporaneous shipments. A one-standard-deviation increase in expertise is associated with a 26.2-percentage-point decline in the probability of

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<sup>31</sup>Full regression results appear in Online Appendix [OA.D.7](#).

Table 2: Probability of Product Drop

|                                            | $Drop_{t:t+5}$       | $Drop_{t:t+5}$       | $Drop_{t:t+5}$       | $Drop_{t:t+5}$       | $Drop_{t:t+5}$       |
|--------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| $\log(PV_{pit})$                           | -0.052***<br>(0.002) |                      |                      | -0.052***<br>(0.002) |                      |
| $q_{ip}$                                   |                      | -1.074***<br>(0.021) |                      |                      | -1.081***<br>(0.022) |
| Young Firm $_{ijt}$                        |                      |                      | -0.075***<br>(0.003) | -0.141***<br>(0.007) | -0.368***<br>(0.022) |
| $\log(PV_{pit}) \times \text{Young}_{ijt}$ |                      |                      |                      | 0.010***<br>(0.001)  |                      |
| $q_{ip} \times \text{Young}_{ijt}$         |                      |                      |                      |                      | 0.497***<br>(0.043)  |
| Observations                               | 1,622,000            | 1,622,000            | 1,622,000            | 1,622,000            | 1,622,000            |
| Fixed effects                              | $pt, i$              | $pt, i$              | $pt, i$              | $pt, i$              | $pt, i$              |
| Controls                                   | Full                 | Full                 | Full                 | Full                 | Full                 |

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

dropping a product over the next five years. These findings indicate that product dropping is systematically related to both current performance and persistent firm-product characteristics, consistent with selection toward high-expertise products.

**Product Dropping over the Firm Life cycle** Lemma 1 shows that lower information-processing ability strengthens the shrinkage of beliefs toward the prior mean. Under the conditions in Proposition 4(i), it also lowers both the product-dropping and search thresholds. Since information-processing ability increases with firm age in the model, we examine how the probability of dropping a product and its relationship with expertise differ between young and mature firms. We first estimate equation (3.41) with the young-firm indicator, omitting product performance and the interaction term. Column 3 of Table 2 shows that young firms are 7.5 percentage points less likely to drop a product over the next five years.<sup>32</sup>

<sup>32</sup>Full regression results with and without firm-level controls appear in Online Appendix OA.D.7.

Two controls are particularly relevant to interpreting age differences in product dropping. Product scope accounts for differences in firms' portfolios: dropping a firm's sole product entails exit unless the firm adds another product. Total assets help account for financial constraints through the role of physical assets as collateral, since financing difficulties may limit firms' ability to replace existing products. Together, these controls help address alternative explanations for age differences in product dropping.

To examine whether selection on product performance also varies with firm age, we estimate the unrestricted specification in equation (3.41), interacting each performance measure with the young-firm indicator. Here,  $\beta_3$  captures the difference between young and mature firms in the relationship between product performance and dropping. Columns 4 and 5 report positive interaction coefficients for both contemporaneous shipments and product-specific expertise. Thus, product dropping is less sensitive to performance among young firms. The estimates in column 5 imply that a one-standard-deviation increase in expertise is associated with a 26.4-percentage-point decline in the dropping probability among mature firms, compared with 14.2 percentage points among young firms. Selection on expertise is therefore approximately 46% weaker among young firms.

These patterns are consistent with the model's learning mechanism, in which less precise information among young firms weakens selection toward high-expertise products. The results also hold when we replace the young-firm indicator with continuous firm age, as reported in Online Appendix Table OA.31.

The product addition results provide complementary evidence. Young firms are 14.8 percentage points more likely to add a product over the next five years. In the sample used to assess addition quality, however, they are 11.2 percentage points less likely to have additions whose average expertise exceeds that of their existing portfolio. Online Appendix OA.D.4 presents the specifications and results. Together, the dropping and addition results suggest that young firms experiment with products but exhibit weaker selection on expertise, consistent with learning frictions early in the firm life cycle.

Table 3: Probability of Product Drop

|                         | $Drop_{t:t+5}$        | $Drop_{t:t+5}$       | $Drop_{t:t+5}$       |
|-------------------------|-----------------------|----------------------|----------------------|
| $q_{ip}$                | -1.055***<br>(0.041)  | -1.127***<br>(0.033) | -1.119***<br>(0.035) |
| $m_{ijt}$               | 0.465***<br>(0.041)   | 0.406***<br>(0.033)  | 0.427***<br>(0.036)  |
| $q_{ip} \times m_{ijt}$ | -1.036***<br>(0.082)  | -0.832***<br>(0.062) | -0.912***<br>(0.068) |
| Observations            | 1,622,000             | 1,622,000            | 1,622,000            |
| Fixed effects           | $pt, i$               | $pt, i$              | $pt, i$              |
| Controls                | Full                  | Full                 | Full                 |
| Measure                 | Structured management | Data availability    | Data use intensity   |

Notes:  $q_{ip}$  denotes normalized product-specific expertise, and  $m_{ijt}$  denotes the measure indicated in each column. Column (1) uses the structured management score of Bloom et al. (2019), column (2) uses data availability (Q27), and column (3) uses data use intensity (Q28). Estimates of the fixed effects and constant are not reported. Standard errors clustered at the five-digit SIC product level appear in parentheses. Observations are unweighted, and observation counts are rounded under Census Bureau disclosure avoidance procedures. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Product Dropping and Information Processing** In the model, management and data availability shape information-processing precision and thereby firms' beliefs about product-specific expertise. We examine whether product selection varies with management practices, data availability, and the intensity of data use. In equation (3.41), we set  $Q_{ipt} = q_{ip}$  and replace the young-firm indicator with  $m_{ijt}$ , which denotes the structured management score of Bloom et al. (2019), data availability (Q27), or data use intensity (Q28). Online Appendix OA.D.7 presents the specification and full results.

Table 3 reports negative and statistically significant interaction coefficients across all three measures. Thus, the dropping probability declines more sharply with expertise among firms with higher management scores, greater data availability, or more intensive data use. These results complement the firm-age evidence and are consistent with more precise information strengthening selection toward high-expertise products.

### 3.2.4 Product Switching and Firm Performance

Having documented how firms select products, we next examine how product additions and drops relate to firm growth and performance.

Online Appendix [OA.D.5](#) shows that product additions are positively associated with growth in real value added, employment, total factor productivity, and labor productivity, while product drops are negatively associated with growth. Under the additive specification, firms that add and drop products have higher predicted growth than firms that do neither. The implied differences are 0.076 log points for real value added, 0.071 for employment, and 0.049 for total factor productivity and labor productivity.

**Decomposition of Firm-Level Growth in Output and Product Expertise** To examine the role of within-firm reallocation through product switching, we conduct an accounting decomposition of firm-level growth. We measure growth as the change in the shipment-weighted average of log real product shipments and similarly examine changes in shipment-weighted product-specific expertise. The decomposition distinguishes within-product changes, shifts in output shares among continuing products, their interaction, and contributions from product additions and drops. Online Appendix [OA.D.6](#) provides the formula and detailed results.

Table 4 presents the normalized contributions, which we refer to as shares.<sup>33</sup> Panel A shows that growth in our shipment-based measure is primarily driven by within-product growth and new product additions. For young firms, however, new product additions play a more prominent role than within-product growth. The combined contribution of additions and drops is approximately 56% for young firms and 44% for mature firms, or 28% higher for young firms. Panel B shows that product additions also account for a larger share of the increase in shipment-weighted expertise among young firms.

Taken together, these findings connect product switching to firm performance. The

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<sup>33</sup>Average five-year growth in the shipment-based measure is 0.112, 0.662, and 0.104 log points for all, young, and mature firms, respectively. The corresponding changes in the shipment-weighted average of  $\theta_{ip}$  are 0.017, 0.130, and 0.016.

Table 4: Product Switching and Firm Growth

|                                                                        | All firms | Young firms | Mature firms |
|------------------------------------------------------------------------|-----------|-------------|--------------|
| <i>Panel A: Growth in shipment-weighted log product shipments</i>      |           |             |              |
| Within-product growth                                                  | 0.543     | 0.444       | 0.545        |
| Between-product reallocation                                           | 0.008     | 0.002       | 0.009        |
| Cross term                                                             | 0.009     | -0.005      | 0.010        |
| Product additions                                                      | 0.424     | 0.554       | 0.422        |
| Product drops                                                          | 0.015     | 0.005       | 0.015        |
| <i>Panel B: Growth in shipment-weighted product-specific expertise</i> |           |             |              |
| Between-product reallocation                                           | 0.318     | 0.126       | 0.320        |
| Product additions                                                      | 0.574     | 0.853       | 0.572        |
| Product drops                                                          | 0.107     | 0.022       | 0.108        |

*Notes:* Entries are normalized contributions to five-year changes in the shipment-weighted average of log real product shipments (Panel A) and estimated product-specific expertise,  $\theta_{ip}$  (Panel B). Young firms are aged five or less; mature firms are aged six or more. The drop row includes the minus sign in Online Appendix equation (OA.D.3). Within and cross terms are zero for time-invariant expertise. Shares may not sum to one after rounding. Online Appendix OA.D.6 provides the decomposition and results.

greater role of product additions among young firms, alongside their weaker selection on expertise, is consistent with the model’s learning mechanism: firms experiment with products, while limited information-processing ability constrains selection toward high-expertise products early in the firm life cycle.

### 3.3 Robustness Tests

We assess whether age differences in product selection are sensitive to expertise estimation, normalization, statistical inference, and firm exit. Online Appendix OA.D.9 reports results. Observation histories may affect the precision of estimated expertise, particularly for young firms. We therefore re-estimate expertise using only the first two observations per firm-product pair and use 5-, 10-, and 15-year rolling estimation windows. The interaction between expertise and the young-firm indicator remains positive and statistically significant across these specifications, although its magnitude varies.

Expertise estimates may also capture firm-level determinants of shipments not fully accounted for by baseline controls. We vary these controls, including a specification

without them, for which the positive age interaction persists. The age difference in selection also persists under standardization and percentile rankings within industry-year or product-year cells. With bootstrapped standard errors, the expertise coefficient remains negative and the young-firm interaction positive, with both statistically significant. Finally, the main age-related selection pattern persists after excluding firms that subsequently exit. Online Appendix [OA.D.11](#) discusses alternative specifications with firm-year fixed effects and an AKM-style residual-based measure of expertise.

## 4 Quantitative Analysis

We quantify how imperfect learning shapes selection within and across firms and aggregate outcomes. We extend the baseline model to allow for persistent demand shocks and calibrate it to U.S. manufacturing data. After assessing model fit and external validation, we examine how product selection shapes firm continuation and growth and how productive potential predicts future outcomes beyond current output. We then evaluate a managerial-investment subsidy costing 0.1% of baseline consumption expenditure. The subsidy brings product-switching and exit decisions closer to their full-information counterparts, increases cumulative output despite lower firm continuation, and raises stationary consumption by 0.54% after accounting for its resource cost. An accounting decomposition relates the aggregate gain to changes in firm mass, composition, and within-group performance.

### 4.1 Persistent Product-Market Shocks

For the quantitative analysis, we allow the transitory component of product-market demand to be persistent. We retain  $\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt}$  and assume

$$\varepsilon_{ip,t+1} \mid \varepsilon_{ipt} \sim N(\rho \varepsilon_{ipt}, (1 - \rho^2) \sigma_\varepsilon^2), \quad 0 \leq \rho < 1. \quad (4.42)$$

Stationary variance remains  $\sigma_\varepsilon^2$ , and the additional signal (2.2) is unchanged. Persistence reduces the independent information in successive market outcomes: each observation after the first contributes precision  $(1 - \rho)/[(1 + \rho)\sigma_\varepsilon^2]$  about permanent expertise.

Current realized demand also predicts future demand, so the product-line state additionally retains  $\log \theta_{ipt}$ . For positive tenure, the demand term becomes

$$\begin{aligned} \log \mathcal{B}_{it}(\hat{\theta}_{ipt}, \log \theta_{ipt}, \tau_{ipt}) &= \frac{\eta - 1}{\eta} \left[ \rho \log \theta_{ipt} + (1 - \rho) \hat{\theta}_{ipt} \right] \\ &\quad + \frac{1}{2} \left( \frac{\eta - 1}{\eta} \right)^2 \left[ (1 - \rho)^2 \hat{\sigma}_{ipt}^2 + (1 - \rho^2) \sigma_\varepsilon^2 \right]. \end{aligned} \quad (4.43)$$

New products retain the common prior and initial demand term  $\mathcal{B}^0$ . Production and product-switching decisions use the revised demand term and Bayesian transition, with the same timing, entry, and exit conventions. Online Appendix OA.B.1 provides the full equations. Setting  $\rho = 0$  recovers the baseline model.

## 4.2 Parameterization

Among potential entrants, fundamental type  $\varphi_i$  and managerial ability  $m_i$  are jointly lognormal:

$$\begin{pmatrix} \log \varphi_i \\ \log m_i \end{pmatrix} \sim N \left( \begin{pmatrix} 0 \\ \log m_0 \end{pmatrix}, \begin{pmatrix} \sigma_\varphi^2 & \rho_{m\varphi} \sigma_\varphi \sigma_m \\ \rho_{m\varphi} \sigma_\varphi \sigma_m & \sigma_m^2 \end{pmatrix} \right). \quad (4.44)$$

The parameters  $\sigma_\varphi$  and  $\sigma_m$  govern dispersion in log productivity and log managerial ability, while  $\rho_{m\varphi}$  governs their correlation before endogenous selection.

Productivity and potential scope are

$$z_i = z_0 \varphi_i, \quad n_i = \max \{1, \lfloor n_0 \varphi_i^{\alpha_n} \rfloor\}. \quad (4.45)$$

Here  $n_0 > 0$  scales potential scope and  $\alpha_n \geq 0$  governs its responsiveness to firm type.<sup>34</sup>

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<sup>34</sup>We choose  $\alpha_n$  so that the maximum  $n_i$  on the computational grid is well below  $N/2$ .

Realized scope depends on product-switching decisions.

We proxy accumulated data by

$$D_{it} = D_0 + n_i(a_{it} + 1), \quad a_{it} > 1. \quad (4.46)$$

At age one,  $D_{it} = D_0 + a_{it} + 1$  reflects data from the initial product. Subsequently, potential scope proxies for the scale of data generation, while  $D_0 \geq 0$  captures other data sources. The data path is determined by firm age and permanent type.

Information-processing ability combines managerial ability with a CES aggregator of experience and data:

$$\psi_{it} = m_i^{\gamma_m} \left\{ 1 + (\gamma_a + \gamma_D) \left[ \frac{\gamma_a}{\gamma_a + \gamma_D} [\log(1 + a_{it})]^{\frac{\kappa-1}{\kappa}} + \frac{\gamma_D}{\gamma_a + \gamma_D} [\log D_{it}]^{\frac{\kappa-1}{\kappa}} \right]^{\frac{\kappa}{\kappa-1}} \right\}. \quad (4.47)$$

The parameters  $\gamma_a, \gamma_D > 0$  govern the contributions of experience and data, and  $\kappa > 0$  governs their substitutability. Managerial ability scales information processing with elasticity  $\gamma_m > 0$ . At age zero,  $\psi_{it} = m_i^{\gamma_m}$ . The logarithmic inputs allow the contributions of experience and data to increase at diminishing rates.

### 4.3 Solution Algorithm

The product-line state is  $(\hat{\theta}, \log \theta, \tau, a, \varphi, m)$ , with two time-varying continuous states,  $(\hat{\theta}, \log \theta)$ . We normalize the wage to one and initialize  $m_0, z_0, P^\eta C$ , and the value functions  $V^{ns}, V^{cs}$ , and  $V^{ds}$ . We then: (i) construct  $z_i, m_i, n_i, D_{it}, \psi_{it}$ , the posterior-variance schedule, the joint belief and demand transition, and  $\mathcal{B}(\xi)$ ; (ii) solve the product-line Bellman equations and associated policies to convergence; (iii) compute the expected value of entry and the entry cutoff  $\underline{\varphi}^e$ ; (iv) compute the stationary distribution of active and vacant product lines per unit mass of operating entrants, accounting for product switching, endogenous and exogenous exit, search outcomes, and entry; and (v) up-

Table 5: Internally Calibrated Parameters and Target Moments

| Par.                 | Description                   | Value   | Target moment               | Data  | Model |
|----------------------|-------------------------------|---------|-----------------------------|-------|-------|
| $\sigma_\varepsilon$ | Demand shock SD               | 0.890   | SD log TVS                  | 1.54  | 2.73  |
| $\alpha_z^e$         | Productivity investment cost  | 3.47e−6 | Mean log LP                 | 3.80  | 3.69  |
| $\sigma_\varphi$     | SD log productivity           | 0.760   | SD log LP                   | 0.79  | 0.38  |
| $\alpha_m^e$         | Management investment cost    | 3.17e−2 | Mean $Q28$                  | 0.43  | 0.62  |
| $\sigma_m$           | SD log management             | 0.250   | SD $Q28$                    | 0.25  | 0.30  |
| $\rho_{m\varphi}$    | Management–productivity corr. | 0.300   | Corr( $Q28, \log z$ )       | 0.17  | 0.16  |
| $n_0$                | Product scope scale           | 0.053   | Mean product scope          | 2.19  | 2.20  |
| $\alpha_n$           | Product scope curvature       | 2.371   | SD product scope            | 2.89  | 1.90  |
| $\gamma_m$           | Management elasticity         | 8.000   | Drop reg.: $q \times Q28$   | −0.91 | −0.62 |
| $\gamma_a$           | Experience weight             | 0.300   | Drop reg.: $q \times Young$ | 0.50  | 0.17  |
| $\gamma_D$           | Data weight                   | 0.200   | Drop reg.: $q \times Q27$   | −0.83 | −0.48 |
| $f^o$                | Operating fixed cost          | 0.002   | Share dropping              | 0.64  | 0.53  |
| $f^s$                | Search fixed cost             | 0.009   | Share adding                | 0.34  | 0.53  |
| $x$                  | Search success probability    | 0.335   | No. products added          | 1.27  | 1.34  |
| $\bar{f}^e$          | Fixed entry cost              | 0.033   | Startup rate (%)            | 7.48  | 7.60  |

Notes: SD denotes standard deviation; TVS, total value of shipments; and LP, labor productivity. Product scope is realized product scope.  $q$  is the normalized expertise measure and *Young* is the young-firm indicator used in the empirical product-drop regression. Dropping and adding shares refer to firms and are fractions; the annual startup rate is in percent. The implied capability levels and total startup resource cost are  $m_0 = 0.940$ ,  $z_0 = 7.300$ . Values are rounded.

date  $P^n C$  to satisfy free entry, repeating Steps (ii)–(iv) after each update.

Once free entry holds, we recover the entrant mass  $\mathcal{M}^e$  from labor-market clearing and scale the stationary distribution accordingly. We then recover  $P$  and  $C$  consistent with the equilibrium  $P^n C$ . Finally, we update  $(m_0, z_0)$  using the pre-entry capability choice in (2.24), taking aggregate equilibrium objects and inherited capabilities as given. We repeat the equilibrium and capability updates until  $(m_0, z_0)$  and all equilibrium objects jointly converge.

## 4.4 Calibration

We set 12 parameters using external evidence or benchmark values and jointly calibrate 15 parameters to match empirical moments in Table 5. One period represents one year.

We normalize  $\bar{L} = 1$ , set  $D_0 = 0$ , and use  $N = 1,400$  product markets, corresponding to five-digit SIC products. We set  $\beta = 0.966$ , based on an average ten-year real interest rate of 3.54% over 1982–2007.<sup>35</sup> The substitution elasticities are  $\eta = 6.6$  within product markets and  $\nu = 4.0$  across markets, following [Broda and Weinstein \(2006\)](#). We set  $\delta = 0.039$  to the average annual exit rate of manufacturing firms aged 26 or older in the BDS over 2004–2007.<sup>36</sup> We set  $\theta_0 = -0.010$  and  $\sigma_0 = 0.2125$ , using the mean and standard deviation of the firm-product fixed effects estimated from the CMF, each divided by  $\eta - 1 = 5.6$ .<sup>37</sup> The remaining predetermined parameters are demand-shock persistence  $\rho = 0.8$ , the experience–data substitution elasticity  $\kappa = 1.5$ , entry-investment curvature  $\chi = 6$ , and inter-cohort capability transferability  $(\zeta_m, \zeta_z) = (0.15, 0.05)$ .<sup>38</sup>

Table 5 reports fifteen internally calibrated parameters and target moments computed for the CMF sample used in the empirical analysis. The parameters are calibrated jointly; the row associations indicate the principal moments informing each parameter. Following the MOPS scoring procedure, we discretize the model counterparts of data availability ( $Q27$ ) and data-driven decision making ( $Q28$ ) into five categories from zero to one in increments of 0.25. Product-switching moments use five-year intervals.

The value of shipments, productivity, and  $Q28$  moments discipline demand and firm heterogeneity and capability-investment costs, while product-scope moments inform  $(n_0, \alpha_n)$ . The interactions of normalized expertise  $q$  with  $Q28$ , young-firm status, and  $Q27$  in the product-drop regression discipline  $(\gamma_m, \gamma_a, \gamma_D)$ . Product dropping, additions, and the startup rate inform  $(f^o, f^s, x, \bar{f}^e)$ .

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<sup>35</sup>Federal Reserve Bank of Cleveland, [10-Year Real Interest Rate](#), retrieved from FRED.

<sup>36</sup>We use this shorter period because firms with left-censored birth years can by then be unambiguously classified as age 26 or older.

<sup>37</sup>See Appendix B.2 for this normalization.

<sup>38</sup>We set  $\chi = 6 > \eta - 1 = 5.6$  so investment costs eventually dominate productivity returns. A common exponent for both capabilities economizes on parameters, with  $\alpha_m^e$  and  $\alpha_z^e$  capturing differences in investment costs.

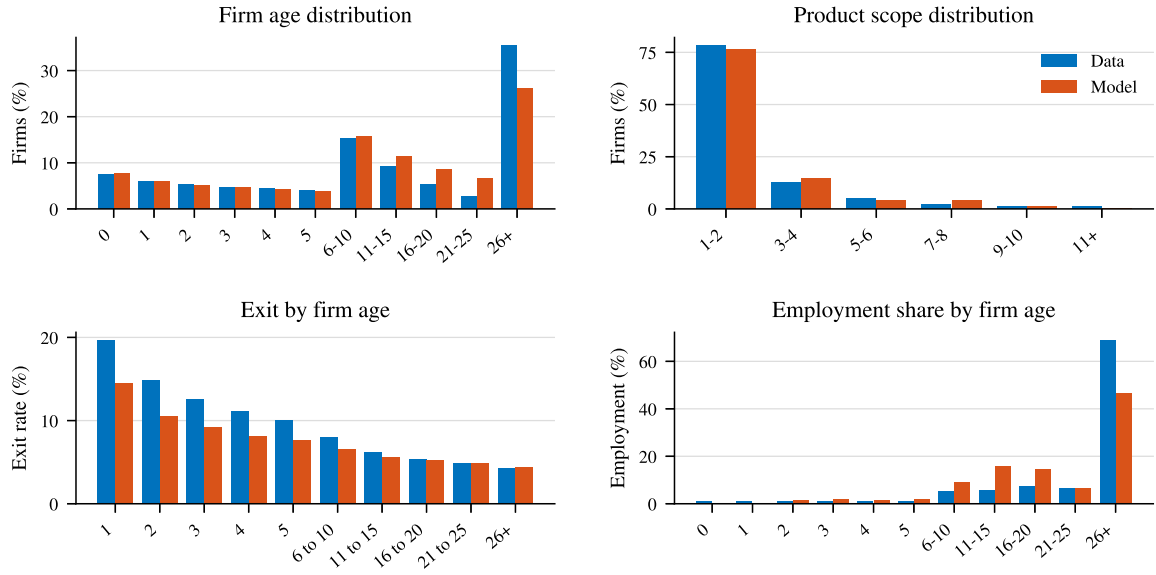


Figure 3: Firm distributions: model and data

Notes: Product scope is the number of currently produced products. The age-26+ group includes left-censored firms in the data.

## 4.5 Model Fit and External Validation

The model closely matches the startup rate and mean product scope and reproduces the signs of the product-dropping interactions (Table 5). It overstates sales dispersion and understates productivity dispersion and the interaction magnitudes. Figure 3 shows that the model captures the concentration of firms in small product portfolios and declining exit with age, while understating young-firm exit and the firm and employment shares at ages 26 and above.

Untargeted comparisons also support the importance of demand heterogeneity. Demand-related components account for 64.4% of the variance of log deflated sales, compared with the 75.8% firm-appeal contribution in [Hottman et al. \(2016\)](#). Entrants' demand and productivity gaps relative to old continuing firms are -1.39 and -0.05 log points, respectively; the corresponding plant-level estimates in [Foster et al. \(2016\)](#) are -0.89 and 0.02. Thus, both comparisons indicate a substantially larger demand disadvantage.

Applying the employment variance decomposition of [Serk et al. \(2021\)](#) to simulated

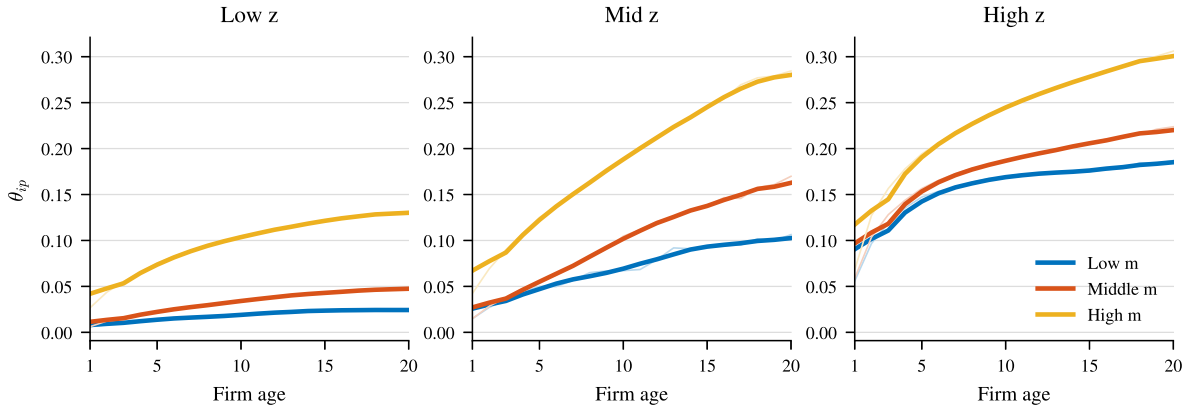


Figure 4: Highest product-specific expertise over the firm life cycle

Notes: Unweighted average across continuing firms of each firm's maximum expertise among currently produced products. Groups are fixed at entry. Faint lines show unsmoothed means; solid lines apply centered five-age smoothing with shorter windows at the endpoints. Profiles reflect both portfolio adjustment and changes in the composition of continuing firms.

firms, we find that ex ante heterogeneity accounts for 67.8% of log-employment variance at age five, compared with 62.1% for manufacturing firms in their data.<sup>39</sup> Online Appendix [OA.C.1](#) reports comparisons, including within-firm sales concentration.

## 4.6 Selection Within and Across Firms

We now use the calibrated model to examine how information processing connects product selection within firms to firm continuation and growth. Proposition 4 provides the mechanism. Under its stated conditions, higher information-processing ability and productivity both encourage search, but have opposite effects on the drop threshold: greater information-processing ability makes firms more selective about retention, whereas higher productivity makes retaining weak products more profitable.

Figure 4 shows that firms' highest product-specific expertise generally rises with age and is higher among firms with greater managerial ability or productivity. These profiles reflect portfolio adjustment within firms and selection among continuing firms. Figure 5 connects these patterns to firm continuation and size. Among high-productivity

<sup>39</sup>The ex ante share declines with age in the data but remains high in the model.

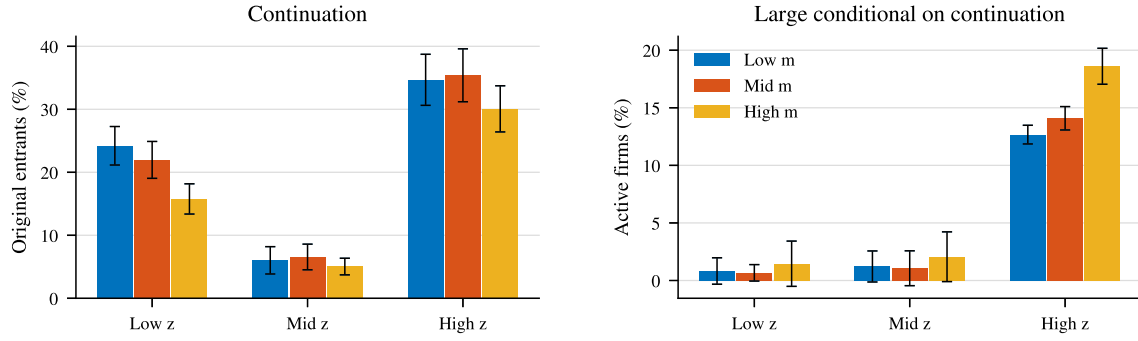


Figure 5: Continuation and large-firm status at age 20

Notes: Probabilities pool firm-age masses within fixed productivity–managerial-ability groups. Continuation is measured relative to original entrants; large-firm status is conditional on continuation. Large firms have employment at or above the age-specific 90th percentile among active firms. Error bars show approximate 95% Monte Carlo intervals. Age-20 statistics pool firm-age masses over ages 18–22.

firms at age 20, those with high managerial ability have a 4.61 percentage-point lower continuation probability but a 5.94 percentage-point higher probability of being large conditional on continuation than those with low managerial ability.<sup>40,41</sup>

These selection patterns highlight the distinction between current output and productive potential. Comparing high- and low- $\varphi_i$  firms with comparable output, the probability of being active and in the top output decile ten years later is higher by 10.14 percentage points at initial age five and 12.37 percentage points at initial age ten.<sup>42</sup> Current output therefore provides an incomplete measure of firms' growth prospects. Online Appendix [OA.C.2](#) describes the comparison and reports additional outcomes.

## 4.7 Policy Experiment: Improving Information Processing

We next quantify how improved information processing affects firm growth, selection within and across firms, and aggregate outcomes. We consider a subsidy to managerial

<sup>40</sup>The joint probability of being active and large is also higher: 5.59% versus 4.39%.

<sup>41</sup>Potential scope also links product selection to firm exit. Low- and middle-productivity firms are restricted to one product, whereas high-productivity firms can drop weak products while retaining others. Middle-productivity firms replace products more intensively than low-productivity firms, increasing their exposure to uncertain products. With no other products to sustain production, dropping a weak product triggers exit, contributing to their lower continuation probability.

<sup>42</sup>In our model,  $\varphi_i$  determines both productivity and potential scope.

investment for entrants above the entry cutoff, provided before initial production. The program uses resources equal to 0.1% of baseline consumption expenditure and raises  $m_0$  from 0.94 to 1.23, holding  $z_0$  and the entry cost fixed. The experiment improves firms' ability to learn without directly changing productivity or product-specific expertise. We first examine how baseline entrants respond at baseline aggregate prices, then account for equilibrium adjustments in prices, entry, and resource allocation.

**Firm Selection and Growth.** We first conduct a paired counterfactual experiment, simulating each baseline entrant with and without the subsidy while holding productive potential, initial product-specific expertise, and random draws fixed.<sup>43</sup> We decompose gains in the cohort's cumulative real sales into production adjustments, product portfolio adjustments, and firm continuation.<sup>44</sup> We aggregate sales across the cohort, deflate them by the baseline product-market price index, and sum from age zero through the displayed age without discounting, reporting totals per original entrant. Sales are zero after exit, capturing changes in continuation without conditioning on survival.

Production adjustments measure the effect of improved information when quantities are reoptimized but baseline product and exit decisions are held fixed.<sup>45</sup> We then allow product switching and exit to adjust. At each age, portfolio adjustments capture the sales effects of product switching among firms producing in both paths. Firm continuation measures sales gained when the subsidy delays exit, net of sales forgone when it accelerates exit. We cumulate these contributions over age and normalize them by cumulative baseline real sales, so they sum to the total percentage change.<sup>46</sup>

Figure 6 shows that improved information processing raises cumulative sales despite lower firm continuation. Continuation at age 20 falls from 20.14% to 17.99%, while cu-

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<sup>43</sup>We exclude changes in entry and equilibrium resource reallocation, which we examine next.

<sup>44</sup>Annual and cumulative responses both indicate gains in real sales, although the contributions of individual margins could vary over the life cycle. We present cumulative responses to summarize the gains accumulated since entry and reduce the influence of year-to-year fluctuations.

<sup>45</sup>This captures the intensive-margin response associated with the output wedge in Proposition 3.

<sup>46</sup>The decomposition is sequential: interactions between improved information and changes in product and exit decisions are assigned to the portfolio-adjustment and firm-continuation components.

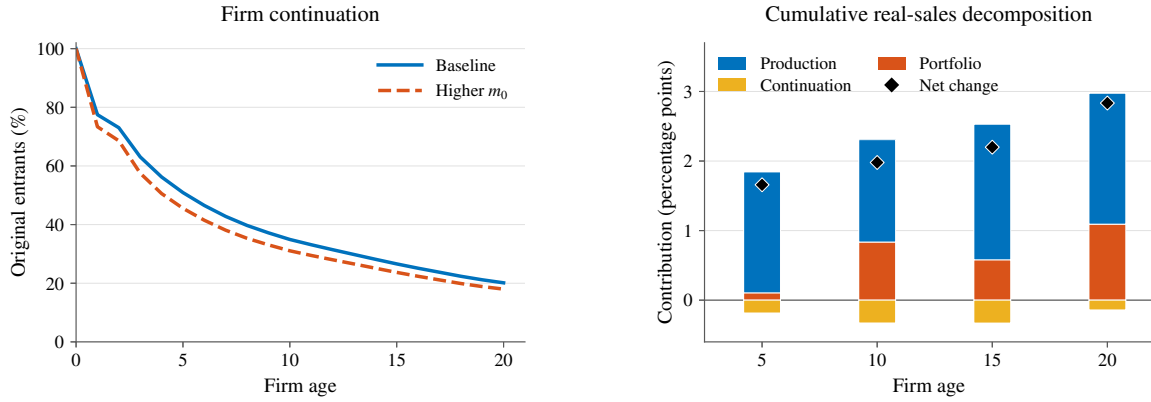


Figure 6: Managerial investment and entrant outcomes

*Notes:* Both panels follow the same baseline entrants at baseline aggregate prices. The left reports continuation among original entrants; the right decomposes cumulative real-sales changes per original entrant. Contributions are percentage points of cumulative baseline real sales; diamonds show their sum. Both panels use centered five-year moving averages.

ulative real sales rise by 2.83%. Production adjustments contribute 1.88 percentage points, portfolio adjustments contribute 1.09 percentage points, and firm continuation subtracts 0.14 percentage points. These components account for 66.4%, 38.6%, and -5.0% of the cumulative sales gain, respectively. The portfolio-adjustment contribution rises from 0.11 percentage points at age five to 1.09 at age 20. Improved information processing therefore raises cumulative sales through both production adjustments within existing portfolios and changes in the products firms operate.

To assess whether portfolio adjustments and changes in continuation reflect better-informed selection, we compare firms' product-switching and exit decisions at each realized state with a full-information benchmark. The benchmark reveals product-specific expertise and reoptimizes current and future decisions.<sup>47</sup> The share of decisions through age 20 that differ from this benchmark falls from 18.22% to 10.72% for product switching and from 8.60% to 5.01% for firm exit (Table 6).

These results quantify how information processing affects firm growth through production adjustments and product selection. Improved information aligns switching and exit decisions more closely with the full-information benchmark. Gains from produc-

<sup>47</sup>Future demand shocks and the expertise of undiscovered products remain unknown.

Table 6: Decisions differing from the full-information benchmark (%)

| Decision          | Ages 0–5 |              | Ages 0–20 |              |
|-------------------|----------|--------------|-----------|--------------|
|                   | Baseline | Higher $m_0$ | Baseline  | Higher $m_0$ |
| Product switching | 21.70    | 13.76        | 18.22     | 10.72        |
| Firm exit         | 12.32    | 7.78         | 8.60      | 5.01         |

*Notes:* Product-switching decisions are continue without search, continue and search, and drop and search. Decisions at each realized state are compared with the full-information benchmark at baseline aggregate prices. The benchmark reveals product-specific expertise and reoptimizes current and future decisions; future shocks and undiscovered expertise remain unknown.

Table 7: Aggregate effects of the managerial investment subsidy

| <i>A. Equilibrium effects</i>                     |                   |
|---------------------------------------------------|-------------------|
| Outcome                                           | Change (%)        |
| Consumption and aggregate productivity            | 0.54              |
| Output                                            | 0.10              |
| Firm mass                                         | -7.67             |
| <i>B. Decomposition of the consumption change</i> |                   |
| Component                                         | Contribution (pp) |
| Firm mass                                         | -8.01             |
| Firm composition                                  | 6.26              |
| Within-group performance                          | 2.29              |
| Total                                             | 0.54              |

*Notes:* Panel B reports an accounting decomposition in percentage points of baseline consumption. Groups are defined by firm age,  $\varphi$ , and baseline managerial ability. Within-group performance measures mean deflated sales per firm and includes changes in firm composition within groups. Interaction terms are allocated symmetrically.

tion and portfolio adjustments more than offset the net sales loss from changes in firm continuation, making continuation alone an incomplete measure of program benefits.

**Selection and Aggregate Outcomes.** The paired experiment isolates production, portfolio, and exit responses for a fixed entrant cohort at baseline aggregate prices. We now examine how these responses affect aggregate outcomes when prices, entry, and resource allocation adjust, comparing stationary equilibria under the same subsidy.

After accounting for the program’s resource cost, consumption rises by 0.54% and

output by 0.10%, while firm mass falls by 7.67% (Table 7).<sup>48</sup>

To connect these aggregate outcomes to changes in selection and performance, we decompose the consumption gain into firm mass, composition across age–type groups, and within-group performance. The first two capture adjustments in the firm population, while within-group performance measures mean deflated sales per firm and incorporates both production and portfolio adjustments.<sup>49</sup> Firm composition and within-group performance contribute 6.26 and 2.29 percentage points, respectively, offsetting the 8.01 percentage-point reduction associated with lower firm mass.

To separate composition and performance from changes in firm mass, we normalize both distributions to baseline firm mass. Under this accounting normalization, aggregate deflated sales rise by 8.90%. Firm composition contributes 6.51 percentage points (73.21%), while within-group performance contributes 2.38 percentage points (26.79%). Online Appendix OA.C.4 details the decomposition and resource accounts. Improved information processing thus raises aggregate consumption through changes in firm composition and performance even though fewer firms operate.

## 4.8 Welfare Under a Restricted Social Planner

We next consider a restricted planner that chooses pre-entry capability investment, production, product switching, startup operation, and entry to maximize household welfare. The planner respects physical feasibility and retains the CE’s value-function restrictions on product switching, startup operation, and entry. Assigned endogenous-exit values create no resources; welfare depends on household consumption. Appendix C formalizes the benchmark.

For fixed  $(m_0, z_0)$  and associated costs, the planner’s admissible stationary allocations coincide with stationary CE allocations. With a common equilibrium selection across

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<sup>48</sup>Aggregate productivity increases by the same percentage as consumption because the labor endowment is fixed. Stationary firm mass reflects both entry and exit.

<sup>49</sup>Within-group performance also reflects equilibrium price adjustments and selection among firms within each group. The accounting components therefore do not map one for one to the margins in the paired experiment.

the two problems, consumption and welfare coincide. Proposition C.1 in Appendix C.2 establishes this replication result and its extension to fixed capability paths.

We are currently computing a stationary solution of the discounted planner problem and will subsequently solve the full sequential problem, accounting for transition dynamics from the baseline CE.<sup>50</sup> In the sequential problem, allowing capability choice weakly raises welfare because the baseline CE path remains feasible (Corollary C.1). These exercises will quantify the welfare implications and associated changes in managerial and productivity investment, entry, and product selection.

## 4.9 Discussion and Policy Implications

Our framework connects learning-driven firm growth and exit, as in Jovanovic (1982), with product selection within firms. Information-processing ability shapes both production and decisions to retain, replace, or drop products as firms learn about their expertise. Managerial investment before entry therefore works through information rather than through physical productivity or product-specific expertise, linking pre-entry capabilities to later production, selection, and growth.

When capabilities are fixed, the restricted planner replicates the equilibrium allocation (Proposition C.1). Within this benchmark, the scope for welfare gains comes from capability investment: entrants do not internalize how their investments benefit future cohorts. Evidence that managerial and entrepreneurial capabilities can be learned and transmitted supports this channel (Giorelli, 2019; Guiso et al., 2021). The externality applies to both managerial and productivity capabilities, and the appropriate investment mix depends on their returns, acquisition costs, and transferability.

Management training, mentoring, and improved access to data can strengthen firms' information-processing ability. Our policy experiment shows that better information raises firm performance through production and portfolio adjustments and increases aggregate consumption, even as fewer firms continue. Entry and survival counts alone

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<sup>50</sup>Both numerical solvers are implemented.

can therefore miss important benefits of capability investment. Policy evaluation should account for changes in production, product selection, and firm composition, alongside the resources required to improve these capabilities.

## 5 Concluding Remarks

In this paper, we develop a model of product switching over the firm life cycle in which firms learn about their product-specific expertise. Information-processing ability depends on managerial ability, experience, and data, so firms learn at different rates even at equal firm-product tenure. Better information makes firms more selective: they drop weak products sooner and search more for replacements. The framework links pre-entry capability investment to learning, portfolio adjustment, firm growth, and exit.

In U.S. manufacturing firm-product data, the probability of dropping a product declines with estimated expertise, but this relationship is weaker among young firms. Selection on expertise is stronger among firms with higher management scores, greater data availability, and more intensive data use. These patterns are consistent with better information strengthening selection toward the products firms are good at.

In our policy experiment, improved information processing raises cumulative sales through production and portfolio adjustments. In general equilibrium, consumption rises after accounting for the program's resource cost, even as fewer firms continue. Product selection within firms is therefore an important channel connecting information processing to firm growth and aggregate outcomes: returns to better information depend on how firms produce, which products they operate, and which firms continue.

When capabilities are fixed, a restricted planner replicates the equilibrium allocation, so the scope for welfare gains lies in pre-entry capability investment and its spillovers across cohorts. Our ongoing work quantifies these gains, including transition dynamics. More broadly, policies toward firms should be judged not only by how many firms survive, but also by how well firms learn what they are good at.

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## A Theory: Proof Sketches and Supporting Derivations

This appendix sketches the proofs of the theoretical results in Section 2, in the order in which they appear. Unless stated otherwise, aggregate prices and quantities are stationary,  $0 < \beta < 1$ ,  $0 \leq \delta < 1$ ,  $0 < x < 1$ ,  $\eta > 1$ , and prior and signal variances are positive and finite. Comparisons condition on firm types and the specified information-processing path, and endogenous-exit payoffs follow the convention in Section 2.3.7.

### A.1 Learning and Bayesian Shrinkage

*Proof sketch of Lemma 1.* Fix true expertise and the observation window, without conditioning on product retention. Taking expectations in (2.3) and using (2.4) gives

$$\mathbb{E}[\hat{\theta}_{ipt} \mid \theta_{ip}] = \log \theta_{ip} + \frac{\hat{\sigma}_{ipt}^2}{\sigma_0^2} (\theta_0 - \log \theta_{ip}), \quad \frac{\partial \hat{\sigma}_{ipt}^2}{\partial \psi_{it}} = -\frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2} < 0. \quad (\text{A.1})$$

Lower precision therefore raises posterior variance and the weight on the prior, so that, on average, firms underestimate  $\log \theta_{ip}$  when it is above  $\theta_0$  and overestimate it when it is below. The same comparison holds for ordered precision histories. Integrating over expertise under the current posterior gives the Gaussian transition in (2.5), with mean  $\hat{\theta}_{ipt}$  and, by the law of total variance, variance  $\hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2$ . Online Appendix OA.A.1 gives the full proof.  $\square$

### A.2 Existence, Uniqueness, and Ordering of Thresholds

*Proof sketch of Proposition 1.* The key step is single crossing in the posterior mean.

*Payoff bound.* Put  $r = \eta - 1$  and  $H = \exp\{r\hat{\theta} + r^2\hat{\sigma}^2/2\}$ . Under retention, Gaussian updating implies  $\mathbb{E}[H' \mid \hat{\theta}, \tau] = H$ ; a replacement resets  $H$  to its finite prior value  $H^0$ . Absolute operating and search payoffs are bounded by  $K(1 + H)$  for a finite  $K$ , so under any policy the expected absolute discounted payoffs after  $T$  periods are bounded by  $K' \sum_{j \geq T} \beta^j (1 + H + jH^0) \rightarrow 0$ . Finite-horizon values therefore converge uniformly

on compact state sets.

*Monotonicity.* At fixed tenure and firm states, couple the posterior-mean transitions using the same normal innovation. A higher current mean then gives a higher next mean with the same transition variance. Backward induction gives continuity and weak monotonicity of  $V$ , and strict monotonicity of  $V^{ns}$  and  $V^{cs}$ ; the payoff bound extends these properties to the infinite horizon.

*Single crossing.* Write  $D_t = V_{it}^{ds}(\emptyset)$ ,  $A_t = V_{it}^{ns}(\hat{\theta}^0, 0)$ , and  $d = \beta(1 - \delta)$ , suppressing common firm states. The branch equations imply

$$\begin{aligned} V_{it}^{cs} - D_t &= (1 - x)(V_{it}^{ns} - dD_{t+1}), \\ V_{it}^{ns} - V_{it}^{cs} &= w_t f^s + x(V_{it}^{ns} - A_t). \end{aligned} \tag{A.2}$$

Both differences are continuous and strictly increasing in the posterior mean. They are negative for sufficiently low beliefs by the low-belief condition in the proposition, and positive for sufficiently high beliefs because operating profits diverge. Each therefore has a unique finite zero. At the drop cutoff, the second difference is negative exactly under (2.18); single crossing then gives the three policy regions. Since  $\mathcal{B}$  is strictly increasing in  $\hat{\theta}$ , the cutoffs carry over to demand units. The first equality also gives the equivalent drop condition

$$\bar{\pi}_{it+1}(\hat{\theta}^{ds}, \tau) + (1 - \delta)\mathbb{E}V_{i,t+1}(\hat{\theta}', \tau + 1) = (1 - \delta)D_{t+1}. \tag{A.3}$$

Online Appendix [OA.A.3.2](#) gives the full proof. □

### A.3 Destination Assignment and Product-Line Separability

Separability rests on two facts: each feasible destination has the same marginal assignment probability, and payoff equivalence removes the dependence of line values on the firm's other lines through endogenous exit.

*Destination assignment.* With  $J$  searching lines and  $N - n_i$  feasible destinations, a

given destination is selected with probability  $J/(N - n_i)$  and assigned to a given line with probability  $1/J$ . Its marginal assignment probability is therefore  $1/(N - n_i)$ , whatever the other lines do. The restriction  $n_i \leq \lfloor N/2 \rfloor$  ensures enough distinct destinations. Symmetric markets and the common-prior reset make all destinations equally valuable. At a startup's first search, only its initial product is excluded. Online Appendix [OA.A.2](#) gives the details.

*Proof sketch of Lemma 2 and Proposition 2.* Conditional on  $(a_{it}, \psi_{it}, \varphi_i, m_i)$ , profits and belief transitions depend only on the focal line's state. If the focal product is retained, the firm reaches search. If it is dropped or the line is vacant, firm continuation and endogenous exit both yield  $V_{it}^{ds}(\emptyset)$ . Other lines' effects on the exit event therefore drop out of the focal line's payoff at every future date, and the assignment argument removes any dependence on their destinations, so line values are separable. Since operating profits and operating and search costs are additive and there is no firm-level fixed cost, firm value is the sum of line values. This holds from age two; the age-one startup uses (2.20). Online Appendix [OA.A.4](#) gives the full proof.  $\square$

## A.4 Distributions and Aggregate Accounting

*Product-line measures.* The exact firm transition and its active-line projection are (2.30) and (2.31); the numerical implementation also tracks vacant lines. A dropped or vacant line reaches search only if another line of the firm remains active. Conditional on firm age and types, let  $q^\emptyset$  be the marginal probability that a line is vacant after drop decisions. Value separability does not imply statistical independence of vacancy indicators. As an approximation, we treat the other  $n_i - 1$  lines' indicators as independent, so a dropped or vacant line reaches search with probability  $1 - p^e$ , where  $p^e = (q^\emptyset)^{n_i - 1}$  and  $p^e = 1$  for  $n_i = 1$ . At age one, continuation is computed directly from retention of the startup's initial product.

Conditional on no exogenous death, a retained product stays in its market after no search or after a failed replacement search, with weights 1 and  $1 - x$ . Successful

searches enter new markets at the common prior; failed searches in vacant lines leave them vacant. All flows are multiplied by  $1 - \delta$ . Averaging over the firm's other excluded products turns each destination probability  $1/(N - n_i)$  into  $1/(N - 1)$ . Operating entrants spread uniformly across markets. Operating costs count producing lines, search costs count actual searches after both exit decisions, and entry costs apply to all entry attempts, including those abandoned before production. Online Appendix [OA.A.5](#) gives the full laws of motion.

*Price index.* Substituting (2.9) into the CES price index and using  $\mathbb{E}_t[\theta_{ip,t+1}^{(\eta-1)/\eta}] = \mathcal{B}(\xi)$  gives

$$P_{pt}^{1-\eta} = \left( \frac{\eta}{\eta-1} w_t \right)^{1-\eta} \int \mathcal{B}(\xi)^\eta z_i^{\eta-1} \mu_{p,t-1}(d\xi). \quad (\text{A.4})$$

This proves (2.36). Combining it with variable labor demand and labor-market clearing gives (2.37). Online Appendix [OA.A.6](#) gives the search-cost expression and the full derivation.

## A.5 Output Wedges

*Proof sketch of Proposition 3.* With known permanent expertise, the predictive demand term is  $\mathcal{B}^{FI} = \theta_{ip}^{(\eta-1)/\eta} \exp\{(\eta-1)^2 \sigma_\varepsilon^2 / (2\eta^2)\}$ . The ratio of optimal quantities is  $(\mathcal{B}/\mathcal{B}^{FI})^\eta$ , which gives (2.38). Write  $\hat{\sigma}^2 \equiv \hat{\sigma}_{ip,t-1}^2$  and  $r = \eta - 1$ . Conditional on true expertise and the precision path, the posterior mean is normal with mean  $\log \theta_{ip} + \hat{\sigma}^2(\theta_0 - \log \theta_{ip})/\sigma_0^2$  and variance  $\hat{\sigma}^2(1 - \hat{\sigma}^2/\sigma_0^2)$ . The normal moment-generating function then gives

$$\log \mathbb{E}[1 - \omega_{ipt} \mid \theta_{ip}] = \frac{r\hat{\sigma}^2}{\sigma_0^2}(\theta_0 - \log \theta_{ip}) + \frac{r^2\hat{\sigma}^2}{2} \left( 1 + \frac{1}{\eta} - \frac{\hat{\sigma}^2}{\sigma_0^2} \right). \quad (\text{A.5})$$

The expected wedge is positive exactly when this expression is negative, which rearranges to (2.39). Lower information-processing ability raises  $\hat{\sigma}^2$  and therefore lowers the cutoff. Online Appendix [OA.A.7](#) gives the full derivation.  $\square$

## A.6 Convexity and the Value of Further Learning

**Lemma A.1** (Value of Further Learning). *At fixed tenure and firm states, the product-line value is continuous, weakly increasing, and convex in the demand term  $\mathcal{B}$ , and its no-search branch is strictly increasing and strictly convex. Holding current  $\mathcal{B}$  and future values fixed, a larger variance of the next belief update raises expected continuation value.*

*Proof sketch.* Let  $\alpha = (\eta - 1)/\eta$  and let  $v_t^\tau = \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2$  be the variance of the next belief update. Gaussian updating implies the retained-product transition

$$\mathcal{B}' = \mathcal{B} \exp\{\alpha \sqrt{v_t^\tau} u - \alpha^2 v_t^\tau / 2\}, \quad u \sim N(0, 1), \quad \mathbb{E}_t \mathcal{B}' = \mathcal{B}. \quad (\text{A.6})$$

Expected profit is strictly increasing and strictly convex in  $\mathcal{B} > 0$ . Since  $\mathcal{B}'$  is a positive multiple of  $\mathcal{B}$ , taking expectations preserves monotonicity and convexity. So does the maximum over actions, because the continue-and-search value is  $(1 - x)V^{ns}$  plus a term independent of  $\mathcal{B}$ , and the drop-and-search value does not depend on  $\mathcal{B}$ . Backward induction and the payoff bound in Appendix A.2 give the stated properties.

A larger  $v_t^\tau$  is equivalent to multiplying  $\mathcal{B}'$  by an independent lognormal factor with mean one. By conditional Jensen's inequality, this raises the expectation of any convex continuation value, strictly when the value is strictly convex on a set reached with positive probability. Online Appendix OA.A.8 gives the full proof.  $\square$

## A.7 Information Processing, Productivity, and Product Tenure

The proof expresses both thresholds through  $U_t$ , the value of keeping the product rather than holding a vacant line from next period. A lower  $U_t$  raises both thresholds, and a higher net value of a fresh product,  $\Delta_t^V$ , raises the search threshold.

Normalize the wage to one and put  $d = \beta(1 - \delta)$ . Define

$$\begin{aligned} U_t(\mathcal{B}, \tau) &= V_{it}^{ns}(\mathcal{B}, \tau) - dD_{t+1}, \\ \Delta_t^V &= V_{it}^{ns}(\mathcal{B}^0, 0) - dD_{t+1} - f^s/x = \frac{D_t - dD_{t+1}}{x}. \end{aligned} \quad (\text{A.7})$$

Thus  $\Delta_t^V$  is the value of a fresh product relative to a vacant line from next period, net of the search cost per successful replacement. The thresholds satisfy  $U_t(\mathcal{B}_t^{ds}, \tau) = 0$  and  $U_t(\mathcal{B}_t^{cs}, \tau) = \Delta_t^V$ , so a nonempty continue-and-search region requires  $\Delta_t^V > 0$ . Subtracting the vacancy value from the Bellman equation gives

$$U_t(\mathcal{B}, \tau) = \beta \bar{\pi}_{it+1}(\mathcal{B}, z_i) + d\mathbb{E}_t \max\{U_{t+1}(\mathcal{B}', \tau + 1) - x\Delta_{t+1}^V, (1-x)U_{t+1}(\mathcal{B}', \tau + 1), 0\}. \quad (\text{A.8})$$

The maximum is increasing and 1-Lipschitz in  $U$ , decreasing in  $\Delta^V$ , and convex in  $\mathcal{B}'$ . By the payoff bound, truncating this recursion while keeping the actual future values of  $\Delta^V$  has a vanishing error.

*Proof sketch of Proposition 4. Information processing.* Compare model-implied precision paths,  $H$  and  $L$ , at equal tenure, productivity, and aggregate prices. Sufficient conditions are  $v_{t+j}^{\tau+j,H} \leq v_{t+j}^{\tau+j,L}$  and  $\Delta_{t+j}^{V,H} \geq \Delta_{t+j}^{V,L}$  for every  $j \geq 0$ , with a strict current variance inequality. Less remaining learning lowers the expected value of the convex continuation, and better replacement opportunities lower it further. Backward comparison in (A.8) gives  $U_t^H < U_t^L$ . Since  $U_t$  is strictly increasing in  $\mathcal{B}$ , both thresholds rise.

For local comparisons, let  $\lambda$  shift the whole precision path, scaled so that  $\partial\psi_{it}/\partial\lambda = 1$ . Write  $\mathcal{L}_t(b) = -\partial_\lambda U_t(b, \tau)$  and  $s_t^j = U_{t,\mathcal{B}}(\mathcal{B}_t^j, \tau) > 0$  for  $j \in \{ds, cs\}$ . Implicit differentiation gives

$$\partial_\lambda \mathcal{B}_t^{ds} = \frac{\mathcal{L}_t(\mathcal{B}_t^{ds})}{s_t^{ds}}, \quad \partial_\lambda \mathcal{B}_t^{cs} = \frac{\partial_\lambda \Delta_t^V + \mathcal{L}_t(\mathcal{B}_t^{cs})}{s_t^{cs}}. \quad (\text{A.9})$$

Under the local counterparts of the path conditions,  $\partial_\lambda v_{t+j} \leq 0$  and  $\partial_\lambda \Delta_{t+j}^V \geq 0$  for every  $j \geq 0$  with  $\partial_\lambda v_t < 0$ , both  $\mathcal{L}_t$  terms are positive, so a sufficient condition for a wider region is

$$\frac{\partial_\lambda \Delta_t^V}{s_t^{cs}} > \frac{\mathcal{L}_t(\mathcal{B}_t^{ds})}{s_t^{ds}}. \quad (\text{A.10})$$

Here and below, derivatives require differentiability, integrable derivative payoffs, and a vanishing discounted derivative tail.

*Productivity.* Hold the information-processing path, scope, search technology, and

fixed costs fixed. Let  $\mathcal{K}^{ns}$  and  $\mathcal{K}^{ds}$  be the expected discounted operating and search fixed costs under the indicated initial branch and optimal continuation, including later replacements. Since a higher  $z_i$  scales all gross operating payoffs by the same factor, the envelope theorem gives  $z_i V_{z_i}^j / (\eta - 1) = V^j + \mathcal{K}^j$ . Combining this identity with the two indifference conditions gives

$$\begin{aligned}\partial_{z_i} \mathcal{B}_t^{ds} &= \frac{\eta - 1}{z_i} \frac{d\mathcal{K}_{t+1}^{ds} - \mathcal{K}_t^{ns}(\mathcal{B}_t^{ds}, \tau)}{s_t^{ds}}, \\ \partial_{z_i} \mathcal{B}_t^{cs} &= \frac{\eta - 1}{z_i} \frac{f^s/x + \mathcal{K}_t^{ns}(\mathcal{B}^0, 0) - \mathcal{K}_t^{ns}(\mathcal{B}_t^{cs}, \tau)}{s_t^{cs}}.\end{aligned}\tag{A.11}$$

The cost conditions in the proposition make the first derivative negative and the second positive, so the region widens.

*Product tenure.* At the same firm age and information-processing path, one more earlier observation raises posterior precision. It lowers  $v_t^\tau$  at every later date before replacement, because a given signal reduces posterior variance by more when current variance is higher. Replacement opportunities do not change. The same convexity argument gives  $U_t(\mathcal{B}, \tau+1) < U_t(\mathcal{B}, \tau)$ , so both thresholds rise. Online Appendix [OA.A.9](#) gives the full proof and additional sufficient conditions.  $\square$

## A.8 Returns to Pre-Entry Investment

This subsection derives the results in Section 2.6.3. Hold prices and the distribution of normalized types  $(\varphi_i, \tilde{m}_i)$  fixed, with  $z_i = z_0 \varphi_i$  and  $m_i = m_0 \tilde{m}_i$ . Write expected entry value before investment costs as  $\mathcal{E} = \mathcal{G} - \mathcal{K}$ , where  $\mathcal{G}$  is expected discounted revenue less variable production costs and  $\mathcal{K}$  is expected discounted operating and search fixed costs. Both include later replacements and the assigned vacant-line value at endogenous exit.

*Managerial capability.* A firm with higher  $m_0$  can reproduce any lower-precision signal by adding noise and follow the same decisions, so  $\mathcal{E}_{m_0} \geq 0$ .

*Productivity and complementarity.* For a fixed operating and switching policy with optimized quantities, value equals  $z_0^{\eta-1} \mathcal{G}_0 - \mathcal{K}$ , where  $\mathcal{G}_0$  and  $\mathcal{K}$  do not depend on  $z_0$ .

Where the derivatives exist and are integrable, the envelope theorem gives

$$\mathcal{E}_{z_0} = \frac{\eta - 1}{z_0} \mathcal{G} > 0, \quad \mathcal{E}_{z_0 m_0} = \frac{\eta - 1}{z_0} \mathcal{G}_{m_0} = \frac{\eta - 1}{z_0} (\mathcal{E}_{m_0} + \mathcal{K}_{m_0}). \quad (\text{A.12})$$

Boundary terms vanish because payoffs are equal at each decision margin, including startup abandonment and age-one exit. The productivity effect is strict when production occurs with positive probability. Capabilities are therefore locally complementary if and only if  $\mathcal{G}_{m_0} \geq 0$ , and  $\mathcal{E}_{m_0} > 0$  with  $\mathcal{K}_{m_0} \geq 0$  suffices for strict complementarity. Online Appendix [OA.A.11](#) gives the full derivation.

## B Persistent Demand and the Expertise Measure

### B.1 Persistent Product-Market Shocks

Quasi-differencing gives  $\log \theta_{ip,t+1} - \rho \log \theta_{ipt} = (1 - \rho) \log \theta_{ip} + e_{ip,t+1}$ , where  $e_{ip,t+1}$  has variance  $(1 - \rho^2) \sigma_\varepsilon^2$  and is independent of past information and private-signal errors. For positive tenure, normal conjugacy gives

$$\begin{aligned} \hat{\sigma}_{ip,t+1}^{-2} &= \hat{\sigma}_{ipt}^{-2} + \frac{1 - \rho}{(1 + \rho) \sigma_\varepsilon^2} + \frac{\psi_{i,t+1}}{\sigma_\varepsilon^2}, \\ \hat{\theta}_{ip,t+1} &= \hat{\sigma}_{ip,t+1}^2 \left[ \frac{\hat{\theta}_{ipt}}{\hat{\sigma}_{ipt}^2} + \frac{\log \theta_{ip,t+1} - \rho \log \theta_{ipt}}{(1 + \rho) \sigma_\varepsilon^2} + \frac{\psi_{i,t+1} \log s_{ip,t+1}}{\sigma_\varepsilon^2} \right]. \end{aligned} \quad (\text{B.1})$$

A product's first market observation has precision  $1/\sigma_\varepsilon^2$ , because a new product starts with a stationary demand shock, so its first update is the baseline update from  $(\theta_0, \sigma_0^2)$ .

For positive tenure, conditional on current information,  $\log \theta_{ip,t+1}$  is normal with mean  $\rho \log \theta_{ipt} + (1 - \rho) \hat{\theta}_{ipt}$  and variance  $(1 - \rho)^2 \hat{\sigma}_{ipt}^2 + (1 - \rho^2) \sigma_\varepsilon^2$ , which proves (4.43). The pair  $(\hat{\theta}_{ip,t+1}, \log \theta_{ip,t+1})$  is jointly normal, with posterior-mean variance  $\hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2$  and covariance  $(1 - \rho) \hat{\sigma}_{ipt}^2$ . At tenure zero, both means equal  $\theta_0$ , and the posterior-mean variance, covariance, and demand variance are  $\sigma_0^2 - \hat{\sigma}_{ip,t+1}^2$ ,  $\sigma_0^2$ , and  $\sigma_0^2 + \sigma_\varepsilon^2$ . These moments define the transition for the Bellman equations on

the state  $(\hat{\theta}_{ipt}, \log \theta_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i, m_i)$ . Both continuous coordinates are needed, because equal demand terms need not imply equal continuation values. Online Appendix [OA.B.1](#) gives the complete system.

## B.2 Mapping Shipment Fixed Effects to Expertise

Under full information about permanent expertise, its loading in  $\log \mathcal{B}$  is  $(\eta - 1)/\eta$ . In [\(2.11\)](#), its total loading in  $\log$  real shipments is therefore  $(\eta - 1)^2/\eta + (\eta - 1)/\eta = \eta - 1$ . After productivity and aggregate components are removed, this gives the benchmark mapping  $\alpha_{ip}^{PV} \simeq (\eta - 1) \log \theta_{ip}$ . Demand persistence changes shipment dynamics but not this loading. We use this mapping as a benchmark for interpreting the estimated fixed effects; see Online Appendix [OA.B.2](#).

## C Restricted Planner: Formulation and Proof Sketches

Given the initial firm distribution, inherited capabilities, and predetermined quantities, the restricted planner maximizes  $W^{SP} \equiv \sum_{t \geq 0} \beta^t \log \bar{C}_t$  over physically feasible production, product switching, startup operation, entry, and pre-entry investments. It uses firms' information, allocates production efficiently given that information, and keeps the CE valuation rules. From age two, each line is valued independently of its effects on firm exit and other lines and receives its vacant-line value at endogenous exit. At age one, only the initial line receives this value; initially vacant lines receive zero. Startup operation and entry use the same values, which create no resources. Capability technologies and costs are those in [Section 2.4](#); private optimality of capability investment is not imposed. The CE information structure, timing, and physical laws of motion apply, with symmetric markets, positive vacant-line values, and a common rule for breaking ties. Online Appendix [OA.C.6.1](#) gives the formal definition.

## C.1 Conditional Production and Verification

Let  $R_t = \bar{L} - \mathcal{F}_t^o - \mathcal{F}_t^s - \mathcal{F}_t^e > 0$  be labor available for variable production and  $S_t = \int \mathcal{B}(\xi)^\eta z_i^{\eta-1} \mu_{p,t-1}(d\xi) \in (0, \infty)$  the productive index of a symmetric market. Conditional on planning information, an exact law of large numbers gives  $C_{pt}^{(\eta-1)/\eta} = \int \mathcal{B}(\xi) [z_i \ell_{pt}(\xi)]^{(\eta-1)/\eta} \mu_{p,t-1}(d\xi)$ . When quantities can be chosen, concavity and the first-order condition give

$$\ell_{pt}(\xi) = \frac{R_t}{N} \frac{\mathcal{B}(\xi)^\eta z_i^{\eta-1}}{S_t}, \quad C_{pt} = \frac{R_t}{N} S_t^{1/(\eta-1)}. \quad (\text{C.1})$$

An active line's direct marginal contribution to log consumption, divided by the marginal value of labor  $1/R_t$ , is  $\ell_{pt}(\xi)/(\eta-1) - f^o$ . The CE first-order condition gives the same expected profit in labor units,  $\bar{\pi}_t^{SP} = \bar{\pi}_t^{CE}/w_t$ . Online Appendix [OA.C.6.2](#) gives the full derivation.

The planner's decision rules discount labor returns at  $\beta_t^{SP} = \beta R_t/R_{t+1}$  and use the CE line and startup Bellman equations with these operating returns and assigned exit values. Attempted entry  $\mathcal{E}_t \geq 0$  satisfies  $\mathbb{E}_\varphi V_t^e(\varphi) \leq f_t^e$ , with equality if  $\mathcal{E}_t > 0$ , and uses  $f_t^e \mathcal{E}_t$  units of labor. Search and production follow actual firm continuation.

**Lemma C.1** (Verification). *Fix an aggregate path. Assume measurable payoffs and transitions on Borel state spaces, finite-horizon expected absolute returns bounded uniformly over admissible policies, integrable startup values, and a discounted absolute payoff tail that vanishes uniformly over policies. Then the Bellman equations have a finite measurable solution attained by a measurable policy, unique among solutions whose discounted expected terminal values vanish under every admissible policy.*

*Proof sketch.* One-period discount factors multiply to  $\beta^{T-t} R_t/R_T$ . Backward induction and the tail condition give convergence of finite-horizon values and a maximizing policy. Iterating the Bellman inequality shows that any solution satisfying the terminal condition weakly exceeds every policy's return, and a maximizing policy attains it. Online Appendix [OA.C.6.4](#) gives the full argument.  $\square$

For welfare, we also require  $\sum_{t \geq 0} \beta^t |\log \bar{C}_t| < \infty$  and assume that the planner's maximum is attained.

## C.2 Replication and the Value of Capability Choice

**Proposition C.1** (Replication). *Under the maintained restrictions and regularity conditions, fix  $(m_0, z_0)$  and the associated costs. The restricted planner's admissible stationary real allocations then coincide with the stationary CE allocations of the economy with these capabilities fixed, which retains startup operation and free entry but not private capability-investment optimality. For a fixed capability path and costs, the same holds for real allocation paths when both problems start from the same inherited state and predetermined quantities and decentralized values use household intertemporal valuation. Under a common equilibrium selection, consumption and welfare coincide.*

*Proof sketch. Stationary allocations.* With capabilities and their costs fixed, conditional production gives the same real production allocations and  $\bar{\pi}^{SP} = \bar{\pi}^{CE}/w$ . Dividing CE values by the constant wage gives the planner's Bellman, startup-operation, and free-entry conditions, including the age-one rule. Lemma C.1 then gives identical values in labor units and, under a common tie-breaking rule, identical policies. The same physical transitions and resource costs support the same invariant distribution. Conversely, the demand-clearing prices in (2.9) and (2.36) support any admissible stationary planner allocation: multiplying its values by  $w$  recovers the CE decision conditions, and physical feasibility gives market clearing.

*Allocation paths.* Conditional production and symmetry imply  $\bar{P}_t \bar{C}_t = \eta w_t R_t / (\eta - 1)$ , so the household discount factor in labor units is

$$\beta \frac{\bar{P}_t \bar{C}_t}{\bar{P}_{t+1} \bar{C}_{t+1}} \frac{w_{t+1}}{w_t} = \beta \frac{R_t}{R_{t+1}} = \beta_t^{SP}. \quad (\text{C.2})$$

The same normalization and supporting-price argument therefore apply date by date, given common inherited states, predetermined quantities, cohort characteristics, and

terminal conditions. Online Appendix [OA.C.6.5](#) gives the full proof.  $\square$

The result concerns efficiency within the maintained valuation rules, not unrestricted efficiency.

**Corollary C.1** (Value of Capability Choice). *Starting from a stationary CE's firm distribution, inherited capabilities, and predetermined quantities, allowing pre-entry capability choice weakly raises welfare:  $W^{SP} \geq W^{CE}$ .*

*Proof sketch.* Set  $a_t^m = m_0^{CE}$  and  $a_t^z = z_0^{CE}$  at every date. The capability technologies then reproduce the CE capabilities and costs. The CE allocation remains physically feasible and satisfies the valuation rules, so it is available to the planner. Maximizing over a set that contains it gives  $W^{SP} \geq W^{CE}$ .  $\square$